

# **TOSHIBA**

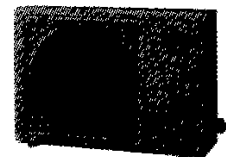
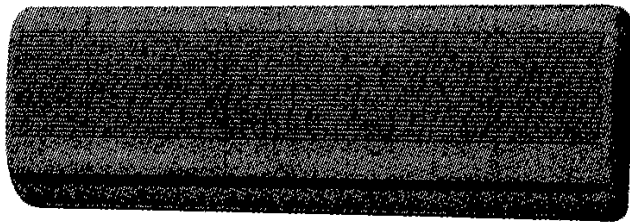
ROOM AIR-CONDITIONER (SPLIT TYPE)

**RAS-07UKH/RAS-07UAH**

**RAS-07UKHX/RAS-07UAHX**

**RAS-09UKH/RAS-09UAH**

**RAS-09UKHX/RAS-09UAHX**



Specifications are subject to change without notice.

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# 1. SPECIFICATIONS

| ITEM                                     | MODEL                                            | RAS-07UKH/RAS-07UAH<br>RAS-07UKHX/RAS-07UAHX |               | RAS-09UKH/RAS-09UAH<br>RAS-09UKHX/RAS-09UAHX |               |
|------------------------------------------|--------------------------------------------------|----------------------------------------------|---------------|----------------------------------------------|---------------|
|                                          |                                                  | COOLING                                      | HEATING       | COOLING                                      | HEATING       |
| Capacity                                 | <sup>*1</sup> kW                                 | 1.8                                          | 1.85          | 2.3                                          | 2.5           |
|                                          | Btu/h                                            | (6,192)                                      | (6,364)       | (7,912)                                      | (8,600)       |
| Power source                             | Phase                                            | Single                                       |               | Single                                       |               |
|                                          | V                                                | 220-240                                      |               | 220-240                                      |               |
|                                          | Hz                                               | 50                                           |               | 50                                           |               |
| Power consumption                        | kW                                               | 0.57                                         | 0.50          | 0.75                                         | 0.68          |
| Power factor                             | %                                                | 98                                           | 98            | 98                                           | 97            |
| Running current                          | A                                                | 2.52                                         | 2.22          | 3.33                                         | 3.05          |
| Starting current                         | A                                                | 10                                           |               | 14                                           |               |
| Moisture removal                         | lit/h                                            | 0.8                                          |               | 1.2                                          |               |
| Noise                                    | Indoor dB                                        | 43/36/31                                     |               | 43/36/31                                     |               |
|                                          | Outdoor dB                                       | 48                                           |               | 48                                           |               |
| Refrigerant                              | Name of refrigerant                              | R-22                                         |               | R-22                                         |               |
|                                          | Rated volume kg (lbs)                            | 0.65 (1.43)                                  |               | 0.67 (1.48)                                  |               |
| Refrigerant control                      |                                                  | Capillary tube                               |               | Capillary tube                               |               |
| Inter-connection pipe                    | Gas side size mm (in)                            | 9.52 (3/8)                                   |               | 9.52 (3/8)                                   |               |
|                                          | Connection type                                  | Flare connection                             |               | Flare connection                             |               |
|                                          | Liquid side size mm (in)                         | 6.35 (1/4)                                   |               | 6.35 (1/4)                                   |               |
|                                          | Connection type                                  | Flare connection                             |               | Flare connection                             |               |
|                                          | Maximum length (of one way) <sup>*2</sup> m (ft) | 10 (32.8)                                    |               | 10 (32.8)                                    |               |
|                                          | Maximum height Indoor unit higher m (ft)         | 5 (16.4)                                     |               | 5 (16.4)                                     |               |
|                                          | Outdoor unit higher m (ft)                       | 5 (16.4)                                     |               | 5 (16.4)                                     |               |
| Condensate drain pipe                    | Outer diameter mm (in)                           | 15 (5/8)                                     |               | 15 (5/8)                                     |               |
| INDOOR UNIT                              |                                                  | RAS-                                         | 07UKH, 07UKHX | 09UKH, 09UKHX                                |               |
| Dimensions                               | Height mm (in)                                   | 265 (10.4)                                   |               | 265 (10.4)                                   |               |
|                                          | Width mm (in)                                    | 790 (31.1)                                   |               | 790 (31.1)                                   |               |
|                                          | Depth mm (in)                                    | 155 (6.1)                                    |               | 155 (6.1)                                    |               |
| Net weight                               | kg (lbs)                                         | 8 (17.6)                                     |               | 8 (17.6)                                     |               |
| Evaporator type                          |                                                  | Finned tube                                  |               | Finned tube                                  |               |
| Indoor fan type                          |                                                  | Cross flow fan                               |               | Cross flow fan                               |               |
| Air volume                               | High fan m <sup>3</sup> /h (CFM)                 | 462 (274)                                    |               | 462 (274)                                    |               |
|                                          | Medium fan m <sup>3</sup> /h (CFM)               | —                                            |               | —                                            |               |
|                                          | Low fan m <sup>3</sup> /h (CFM)                  | —                                            |               | —                                            |               |
| Fan motor output                         | W                                                | 18                                           |               | 18                                           |               |
| Air filter                               |                                                  | Polypropylene net filter (Washable)          |               | Polypropylene net filter (Washable)          |               |
| OUTDOOR UNIT                             |                                                  | RAS-                                         | 07UAH, 07UAHX | 09UAH, 09UAHX                                |               |
| Dimensions                               | Height mm (in)                                   | 530 (20.9)                                   |               | 530 (20.9)                                   |               |
|                                          | Width mm (in)                                    | 770 (30.3)                                   |               | 770 (30.3)                                   |               |
|                                          | Depth mm (in)                                    | 200 (7.9)                                    |               | 200 (7.9)                                    |               |
| Net weight                               | kg (lbs)                                         | 29 (63.9)                                    |               | 29 (63.9)                                    |               |
| Condenser type                           |                                                  | Finned tube                                  |               | Finned tube                                  |               |
| Outdoor fan type                         |                                                  | Propeller                                    |               | Propeller                                    |               |
| Air flow volume                          | m <sup>3</sup> /h (CFM)                          | 1,500 (890)                                  |               | 1,500 (890)                                  |               |
| Fan motor output                         | W                                                | 18                                           |               | 18                                           |               |
| Compressor                               | Model                                            | PH70T1-4C                                    |               | PH94T1-4C                                    |               |
|                                          | Output W                                         | 500                                          |               | 750                                          |               |
| Safety device                            |                                                  | Fuse, Overload relay                         |               | Fuse, Overload relay                         |               |
| Auto louver                              |                                                  | Yes                                          |               | Yes                                          |               |
| Usable outdoor temperature range °C (°F) |                                                  | 21~43 (70~109)                               | -5~21 (23~70) | 21~43 (70~109)                               | -5~21 (23~70) |

Specifications are subject to change without notice.

**Note 1:**

Capacity is based on the following temperature conditions.

|                                    |      | JIS         |             |
|------------------------------------|------|-------------|-------------|
|                                    |      | COOLING     | HEATING     |
| Indoor unit inlet air temperature  | (DB) | 27°C (80°F) | 20°C (68°F) |
|                                    | (WB) | 19°C (66°F) | —           |
| Outdoor unit inlet air temperature | (DB) | 35°C (95°F) | 7°C (45°F)  |
|                                    | (WB) | 24°C (75°F) | 6°C (43°F)  |

**Note 2: CHARGELESS**

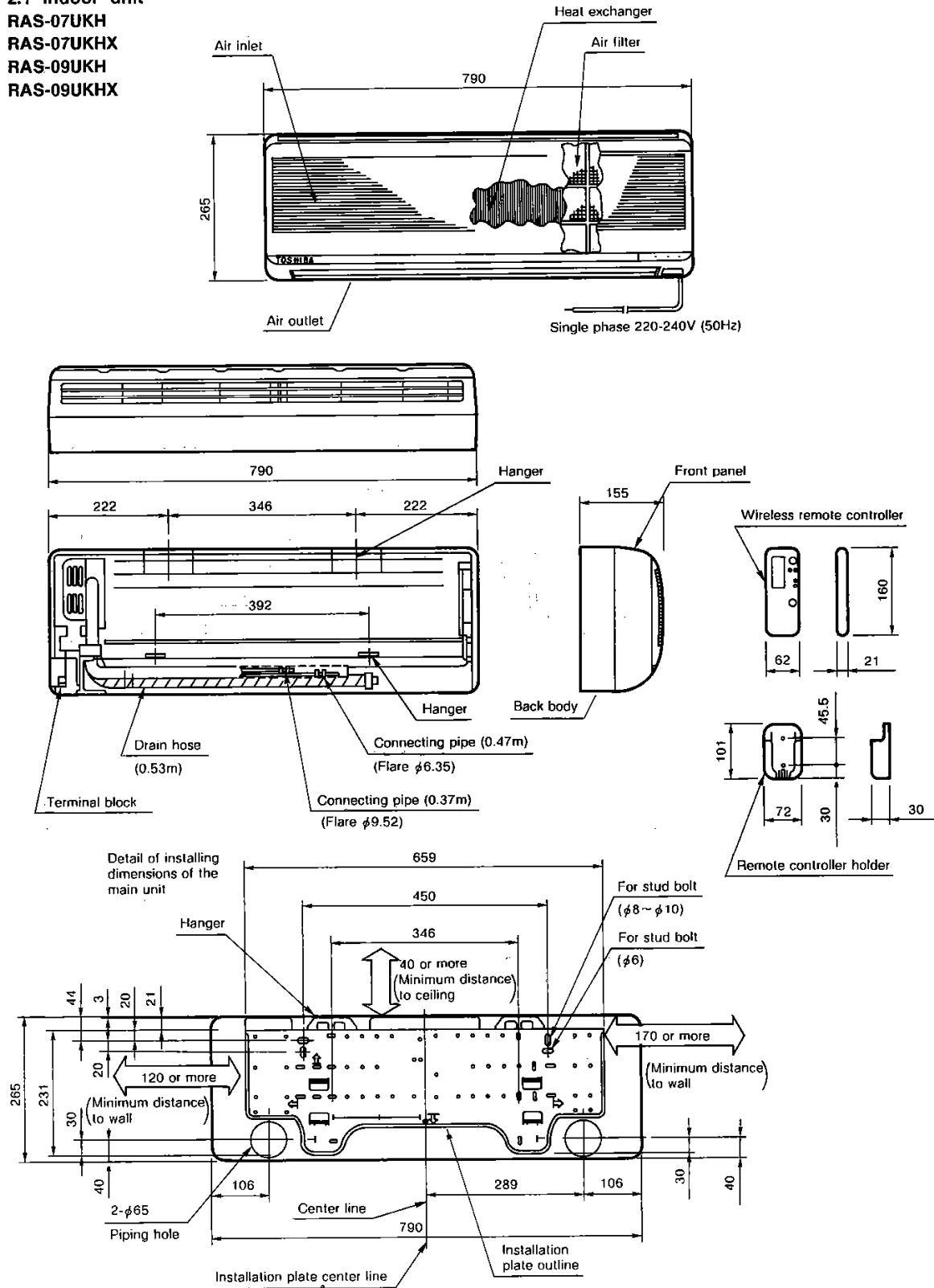
No additional refrigerant required.

This air conditioner accepts a connection piping length of up to 10 m and a head of up to 5 m.

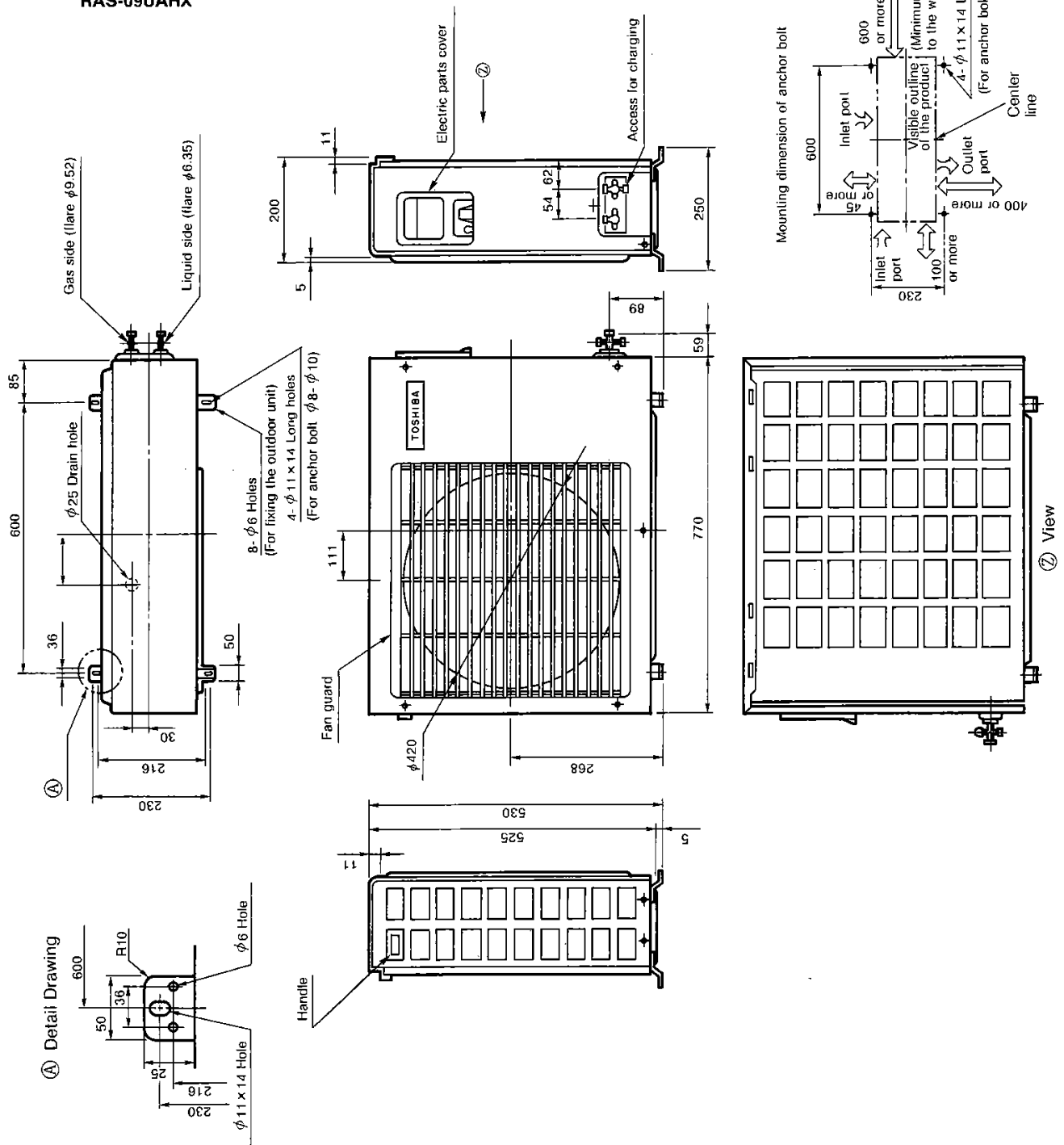
There is no need to add the refrigerant as long as the total length of the connection piping is up to 10 m.

## 2. CONSTRUCTION VIEWS

2.1 Indoor unit  
RAS-07UKH  
RAS-07UKHX  
RAS-09UKH  
RAS-09UKHX



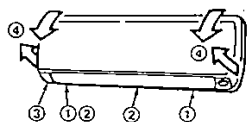
**2.2 Outdoor unit**  
**RAS-07UAH**  
**RAS-07UAHX**  
**RAS-09UAH**  
**RAS-09UAHX**



### 3. CAUTION IN INSTALLATION

#### 3.1 Installation of indoor unit

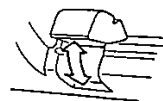
##### a. REMOVING THE FRONT PANEL



Air outlet direction  
louver

- ① Open the screw caps and remove the two screws securing the front panel.
- ② Close the screw caps as before.
- ③ Open the air outlet direction (up - down) louver horizontally by hand.
- ④ Slightly open the lower part of the front panel, then pull the upper part of the front panel toward you to remove it from the body.

##### OPENING THE SCREW CAP

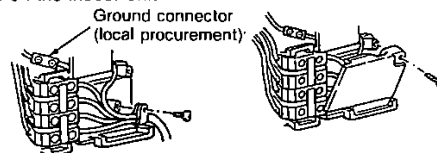
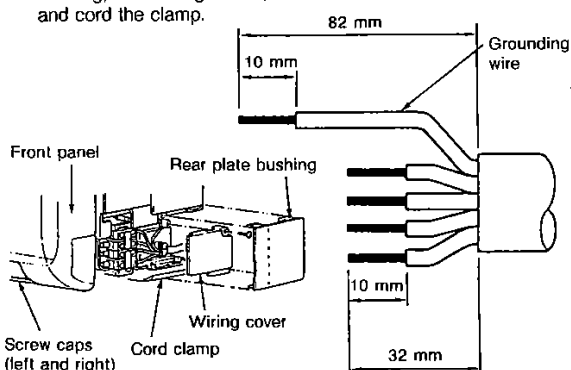


Open it upward by  
putting your finger  
on the edge.

#### CAUTION WHEN CONNECTING THE CONNECTIVE CABLE

When connecting the connective cable, follow the instructions below:

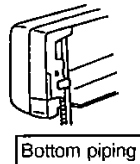
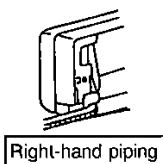
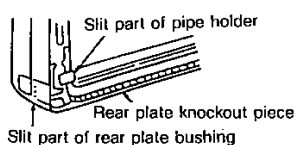
1. • Remove the front panel.  
• Remove the rear plate bushing, the wiring cover, and cord the clamp.
2. Connective cable sheath length to be removed.
3. • Insert the connective cable fully into the terminal block and secure it by screw tightly.  
• Secure the connective cable with the cord clamp.  
• Connect the grounding wire of indoor unit with the grounding wire of connective cable via a ground connector.  
• Fix the wiring cover, the rear plate bushing and the front panel on the indoor unit.



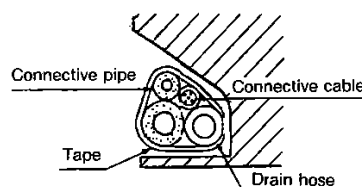
4. • Insert the connective cable fully into the terminal block and secure it by screw tightly.  
• If the connective cable is incompletely inserted, it may be overheated as a result of a defective contact.  
• Use a 1 mm<sup>2</sup> or more connective cable.  
• Secure the cable at its double coated portion with a cord clamp.

##### b. PRECAUTIONS FOR REFRIGERANT PIPING

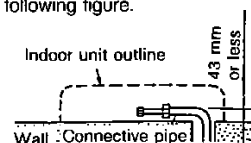
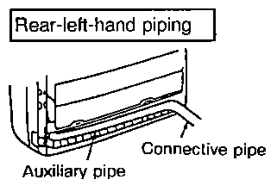
- For the bottom piping, cut out the pipe holder and the slit part on the bottom of the rear plate with a nipper or something.
- For the right-hand piping, cut out the pipe holder and the rear plate bushing at a slit with a nipper or something.



- For right-hand or left-hand piping



- Since the two refrigerant pipes are both used for low pressure during cooling (dry), be sure to thermally insulate the two pipes.
- Do not re-use the nylon cap ( ) at the auxiliary pipe connection of the indoor unit.
- For the rear-left-hand piping, bend the connective pipe as shown in the following figure.



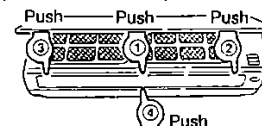
Bend the connective pipe so as to be laid at 43 mm height or less from the wall.

##### c. CAUTION IN MOUNTING THE FRONT PANEL

When the front panel is mounted again after removing, take the following actions:

After fastening the two screws, one each at the left and right of the air outlet, be sure to push the upper center part ①, right end ②, left end ③ and the lower center part ④ of the air outlet and confirm that no gap is left between the front panel and the rear plate.

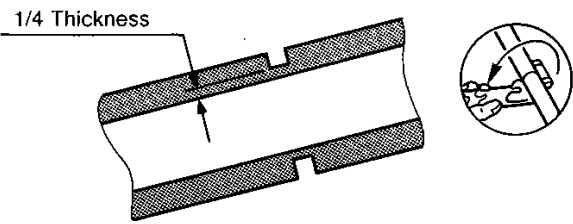
- If cooling (dry) operation is made without pushing the air outlet, dew can be attracted on the front panel surface. In addition, a gap between the front panel and the body will become wider, spoiling the appearance.



### 3.2 Flaring

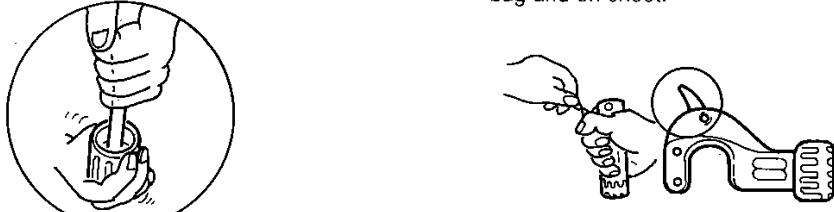
**Pipe Cutting** →

- Cut the pipe little by little, advancing the blade of pipe cutter slowly.
- For fine work, do not advance the blade by force in order to prevent distortion of the pipe, and extra burrs.
- Cut the pipe a little longer than required.



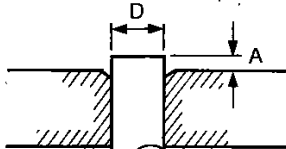
**Burr Removal** →

- Raise burrs with the burr stick. Or remove burrs with the burr remover. (Hold the flaring end down to prevent burrs from dropping inside pipe.)
- Be sure to remove burrs to avoid unevenness on the flare face which will cause gas leaks.
- Cover the pipe end with a polyethylene bag and on sheet.



**Point of Flaring** →

- The exact length of pipe protruding from the face of the flare die is determined by the flaring tool.
- Too long a protrusion of pipe from the flaring tool will result in too long a flare; if this happens, the flare nut will not tighten and the joint will not seat. Too small a protrusion will cause gas leaks.



(The table below shows the use of an imperial die and a rigid die.)

#### CAUTION

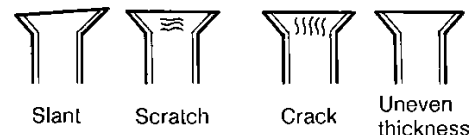
1. Too large a protrusion of pipe results in impossible tightening of the flare nut.
2. Too small a protrusion of pipe will cause gas leaks due to insufficient flare surface.

| Type of Pipe | D (mm) | A (mm)   |       |
|--------------|--------|----------|-------|
|              |        | Imperial | Rigid |
| 1/4"         | 6.35   | 1.3      | 0.7   |
| 3/8"         | 9.52   | 1.6      | 1.0   |

- Fine flaring work shows even brightness on the flared surface and uniform thickness of pipe.
- Unsuccessful flaring work causes gas leaks, therefore rework it.

**Finishing Work** →

- Fix the pipe firmly on the flare die. Match the centers of both the flare die and the flaring punch, and tighten flaring punch fully.

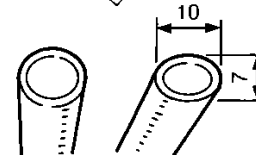
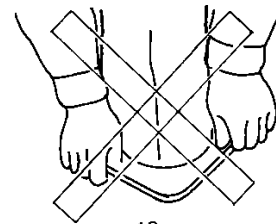
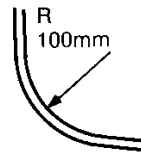
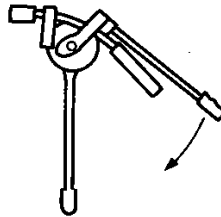


The above faulty flaring will cause gas leaks.

### 3.3 Pipe process points

#### 1. Pipe Bending

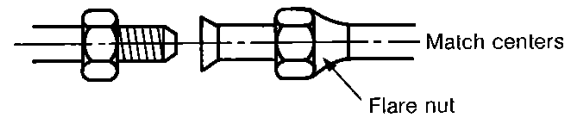
- Carefully bend by hand. -----> • Collapsed pipe cause cooling failure.
- Bend with pipe bender.  
(Since the insulation is wound on, use the bender by sliding the insulation to one side, or cutting it halfway.)
- The radius of the pipe bend should be 100 mm or more.
- Too small a radius causes collapse of pipe.



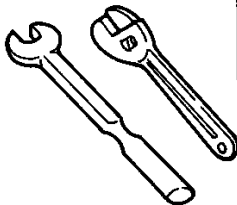
The distortion ratio must be less than 10:7.

#### 2. Pipe Connection

- Connect the flared pipe close to the air -----> • Forcing to tighten the nut without the center axis coincided causes gas leaks.



Flare nut



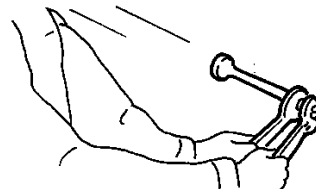
- Use a torque wrench and spanner to tighten or untighten the flare nut. Tightening should be done by finger first, then use a torque wrench. A torque wrench helps to provide adequate torque for tightening. -----> • Insufficient tightening causes gas leaks. Over tightening results in cracks the flared surface.

#### CAUTION

- Do not use the spanner from the beginning of tightening.
- When processing 1/4 inch pipe, tighten lightly with spanner and then tighten additionally by 90~120 degree with torque wrench and spanner.

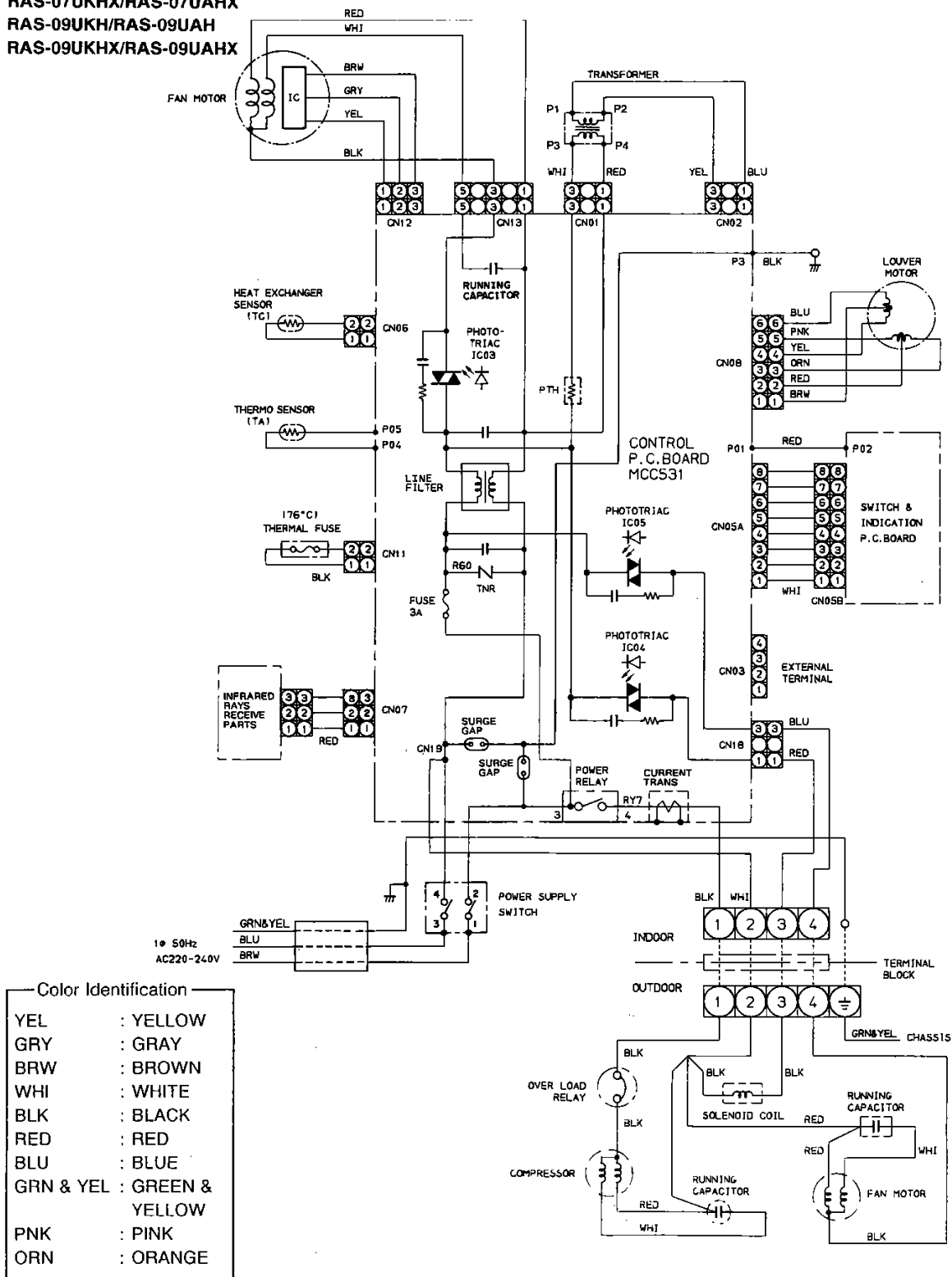
Tightening torque of flare nut (Unit: N·m (kgf·cm))

| Pipe size             | 6.35 mm (1/4") | 9.52 mm (3/8") |
|-----------------------|----------------|----------------|
| Standard torque       | 15.7 (160)     | 29.4 (300)     |
| Additional tightening | 19.6 (200)     | 34.1 (350)     |



#### 4. WIRING DIAGRAM

**RAS-07UKH/RAS-07UAH**  
**RAS-07UKHX/RAS-07UAHX**  
**RAS-09UKH/RAS-09UAH**  
**RAS-09UKHX/RAS-09UAHX**



## 5. SPECIFICATIONS OF ELECTRICAL PARTS

### 5.1 RAS-07UKH, RAS-07UKHX (Indoor unit) RAS-09UKH, RAS-09UKHX

| No. | Parts name                        |                      | Type             | Specifications                                      |                    |                      |
|-----|-----------------------------------|----------------------|------------------|-----------------------------------------------------|--------------------|----------------------|
| 1   | Fan motor (Fan indoor)            |                      | MMF-200-18-2C    | Output (Rated) 18W, 2 pole, 1 phase, 220-240V, 50Hz |                    |                      |
|     |                                   |                      |                  | Winding resistance ( $\Omega$ )<br>(at 20°C)        | Red-Black<br>101.6 | White-Black<br>209.4 |
| 2   | Thermo sensor (TA-sensor)         |                      | (microprocessor) | Adjustable range : 17 ~ 30°C                        |                    |                      |
| 3   | Running capacitor                 | for indoor fan motor | EVN40M105UL1     | 400 WV, 1 $\mu$ F                                   |                    |                      |
| 4   | Transformer                       |                      | ST-8916          | Primary 200 ~ 240V, Secondary 12V                   |                    |                      |
| 5   | Microcomputer                     |                      | TMP47C860N       |                                                     |                    |                      |
| 6   | Power relay                       |                      | G4F-1112TP       | Coil ; 12V 90mA, Rated 250V 15A                     |                    |                      |
| 7   | Heat exchanger sensor (TC-sensor) |                      | (microprocessor) | 10 k $\Omega$ at 25°C                               |                    |                      |

**5.2 RAS-07UAH (Outdoor unit)**  
**RAS-07UAHX**

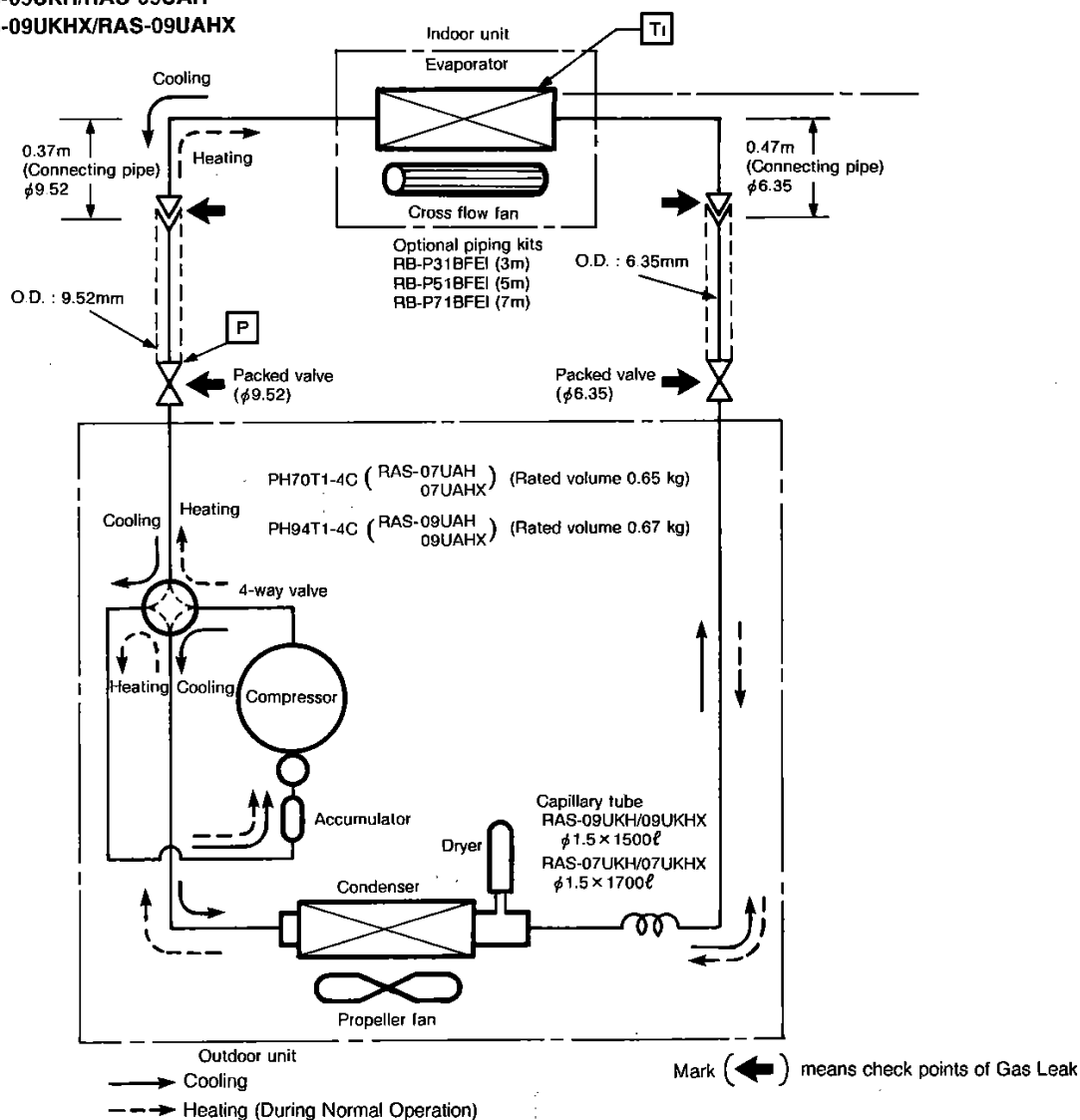
| NO. | PARTS NAME                            | TYPE           | SPECIFICATIONS                                          |           |             |
|-----|---------------------------------------|----------------|---------------------------------------------------------|-----------|-------------|
| 1   | Compressor                            | PH70T1-4C      | Output (Rated) 500 W, 2 pole, 1 phase, 220-240 V, 50 Hz |           |             |
|     |                                       |                | Winding resistance ( $\Omega$ )<br>(at 20°C)            | Red-Black | White-Black |
|     |                                       |                |                                                         | 4.46      | 6.55        |
| 2   | Fan motor (For outdoor)               | UE6-21A5P      | Output (Rated) 18 W, 6 pole, 1 phase, 220/240V, 50 Hz   |           |             |
|     |                                       |                | Winding resistance ( $\Omega$ )<br>(at 20°C)            | Red-Black | White-Black |
|     |                                       |                |                                                         | 366.2     | 368.5       |
| 3   | Running capacitor<br>(For fan motor)  | EEP2G155HQA104 | 400V AC, 1.5 $\mu$ F                                    |           |             |
| 4   | Running capacitor<br>(For compressor) | EAG40M206UF2   | 400V AC, 20 $\mu$ F                                     |           |             |
| 5   | Solenoid coil                         | LB60012        | AC; 220/240V                                            |           |             |
| 6   | Over load relay                       | MSF 40AMM      | AC; 230V 25A                                            |           |             |

**RAS-09UAH (Outdoor unit)**  
**RAS-09UAHX**

| NO. | PARTS NAME                            | TYPE           | SPECIFICATIONS                                          |           |             |
|-----|---------------------------------------|----------------|---------------------------------------------------------|-----------|-------------|
| 1   | Compressor                            | PH94T1-4C      | Output (Rated) 750 W, 2 pole, 1 phase, 220-240 V, 50 Hz |           |             |
|     |                                       |                | Winding resistance ( $\Omega$ )<br>(at 20°C)            | Red-Black | White-Black |
|     |                                       |                |                                                         | 4.53      | 8.73        |
| 2   | Fan motor (For outdoor)               | UE6-21A5P      | Output (Rated) 18 W, 6 pole, 1 phase, 220/240V, 50 Hz   |           |             |
|     |                                       |                | Winding resistance ( $\Omega$ )<br>(at 20°C)            | Red-Black | White-Black |
|     |                                       |                |                                                         | 366.2     | 368.5       |
| 3   | Running capacitor<br>(For fan motor)  | EEP2G155HQA104 | 400V AC, 1.5 $\mu$ F                                    |           |             |
| 4   | Running capacitor<br>(For compressor) | EAG40M206UF2   | 400V AC, 20 $\mu$ F                                     |           |             |
| 5   | Solenoid coil                         | LB60012        | AC; 220/240V                                            |           |             |
| 6   | Over load relay                       | OL-062EQ       | AC; 230V 40A                                            |           |             |

## 6. REFRIGERATING CYCLE DIAGRAM

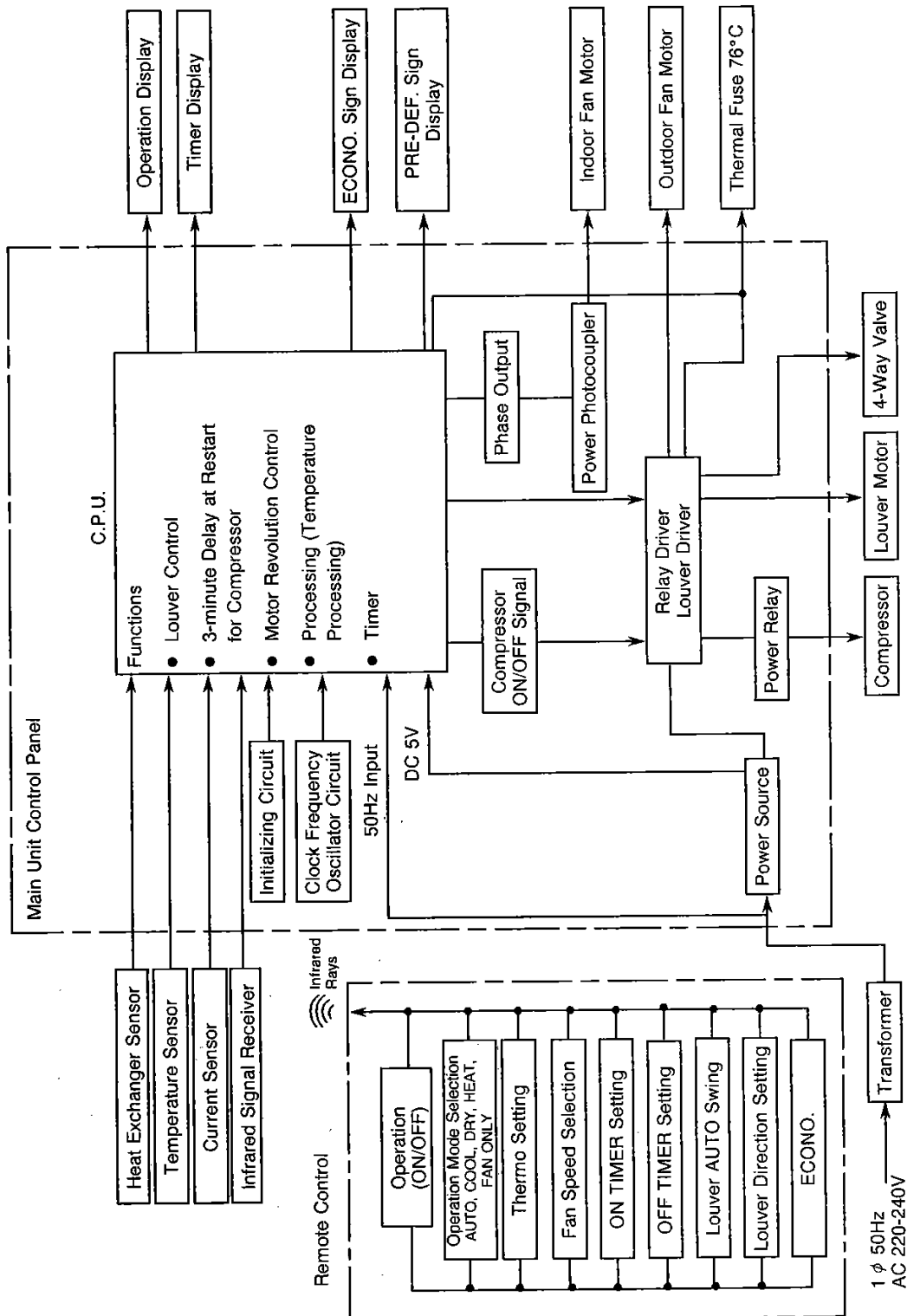
RAS-07UKH/RAS-07UAH  
RAS-07UKHX/RAS-07UAHX  
RAS-09UKH/RAS-09UAH  
RAS-09UKHX/RAS-09UAHX



| Conditions |                  | Standard pressure<br>P (kg/cm <sup>2</sup> G) |             | Surface temp. of heat<br>exchanger interchanging<br>pipe T <sub>1</sub> (°C) |             | Fan<br>speed<br>(Indoor) | Ambient temp. conditions<br>DB/WB<br>(°C) |         |
|------------|------------------|-----------------------------------------------|-------------|------------------------------------------------------------------------------|-------------|--------------------------|-------------------------------------------|---------|
| Model      |                  | RAS-07UKH                                     | RAS-09UKH   | RAS-07UKH                                                                    | RAS-09UKH   |                          | Indoor                                    | Outdoor |
| Heating    | Standard         | 12.6                                          | 14.2        | 34.5                                                                         | 38.5        | High                     | 20/-                                      | 7/6     |
|            | High temperature | 17.1 ~ 19.9                                   | 19.8 ~ 22.8 | 45.5 ~ 52.0                                                                  | 52.0 ~ 59.0 | Low                      | 27/-                                      | 21/15   |
|            | Low temperature  | 10.4                                          | 11.9        | 29.5                                                                         | 34.1        | High                     | 20/-                                      | -5/-5.5 |
|            |                  |                                               |             |                                                                              |             |                          |                                           |         |
| Cooling    | Standard         | 6.5                                           | 6.2         | 12.5                                                                         | 11.4        | High                     | 27/19.0                                   | 35/24   |
|            | High temperature | 6.2                                           | 6.9         | 18.5                                                                         | 14.6        | High                     | 32/23                                     | 43/25.5 |
|            | Low temperature  | 4.6                                           | 4.5         | 3.0                                                                          | 2.4         | Low                      | 21/15                                     | 21/15   |
|            |                  |                                               |             |                                                                              |             |                          |                                           |         |

Note: Measure the heat exchanger temperature at the center of U-bend. (By means of thermistor thermometer)  
During heating overload, the high temperature limit control operation is included.

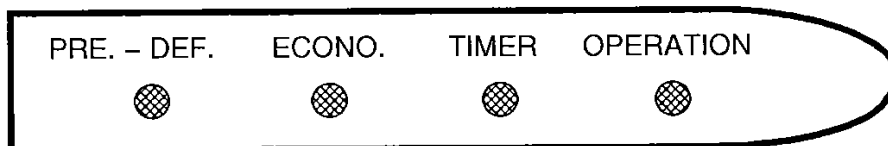
## 7. MICRO-COMPUTER BLOCK DIAGRAM



## 8. OPERATION CIRCUIT DESCRIPTIONS

### NAMES AND FUNCTIONS OF INDICATORS AND CONTROLS ON INDOOR UNIT

#### Display panel

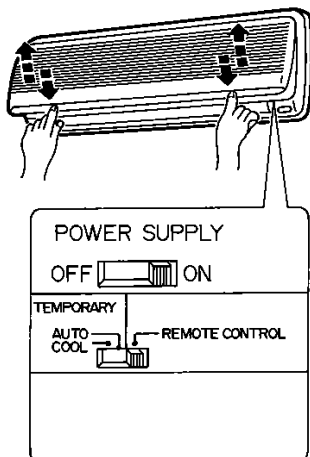


- PRE.-DEF. indicator lamp :: Red LED (lights during PREHEAT or DEFROST operation)
- ECONO. indicator lamp :: Green LED (lights during ECONO. operation)
- TIMER indicator lamp :: Yellow LED (lights during timer operation)
- OPERATION indicator lamp :: Green LED (lights during operation)

**Control panel** Close the panel cover after you set the control switches.

The temporary operation may be used when the remote controller is either lost, or inoperative due to exhausted batteries.

Set the switch to the REMOTE CONTROL position during normal operation.



#### If the TEMPORARY switch is set to AUTO

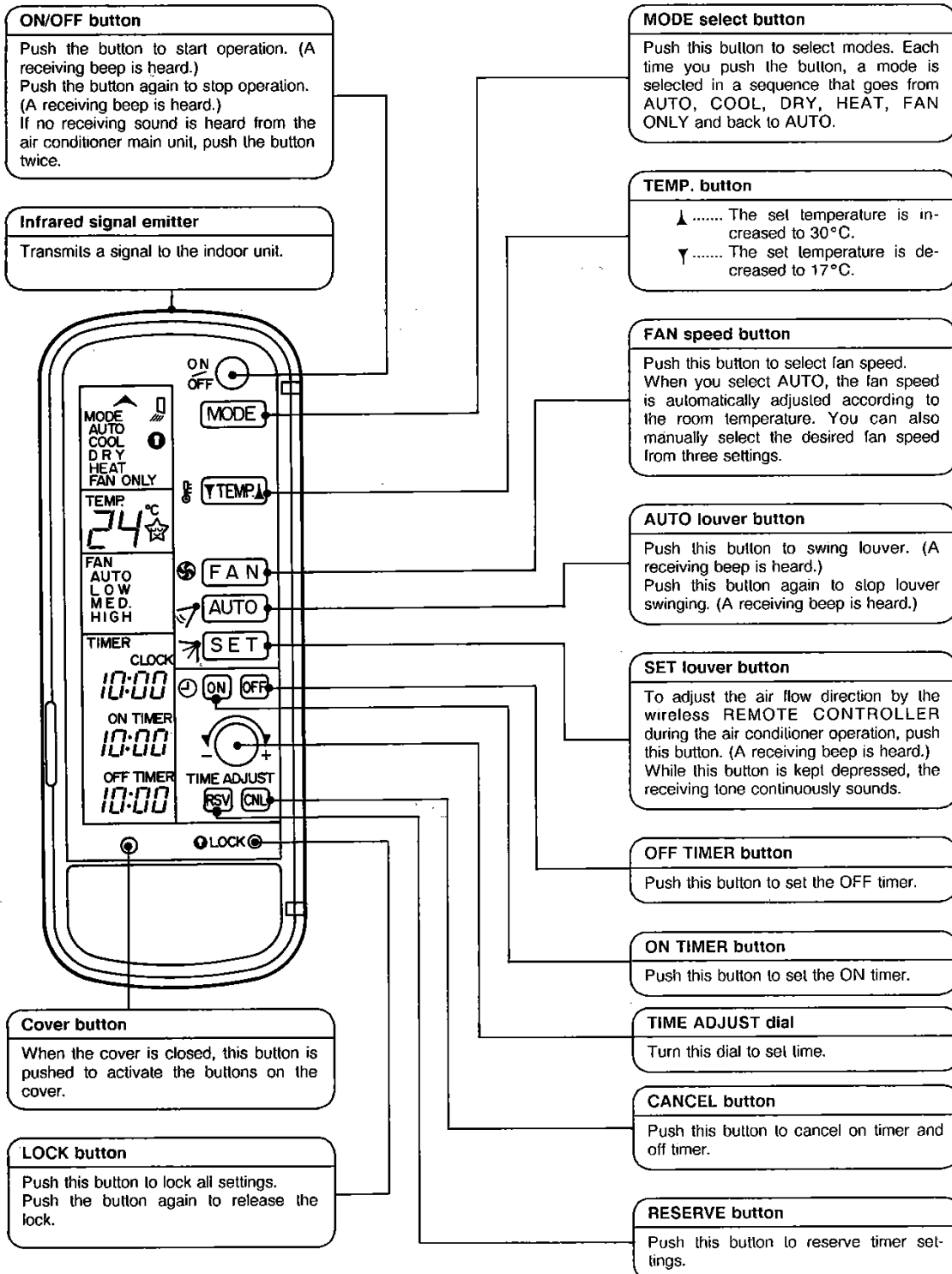
The air conditioner operates by automatically selecting the cooling, dry, fan only mode depending on the room temperature.

#### If the TEMPORARY switch is set to COOL

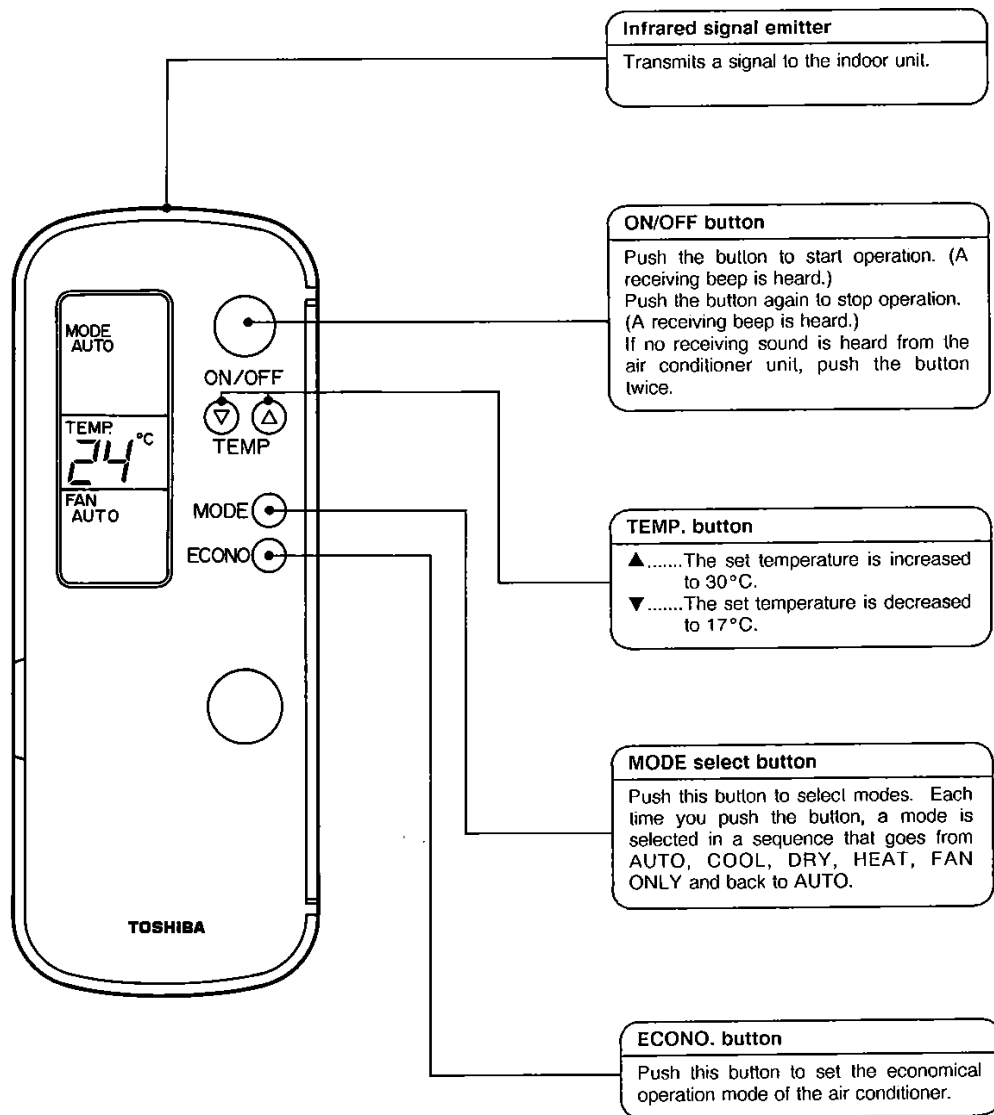
The air conditioner operates in the cooling mode (at a low fan speed).

- If the TEMPORARY switch is set to AUTO or COOL, the remote controller is not usable.

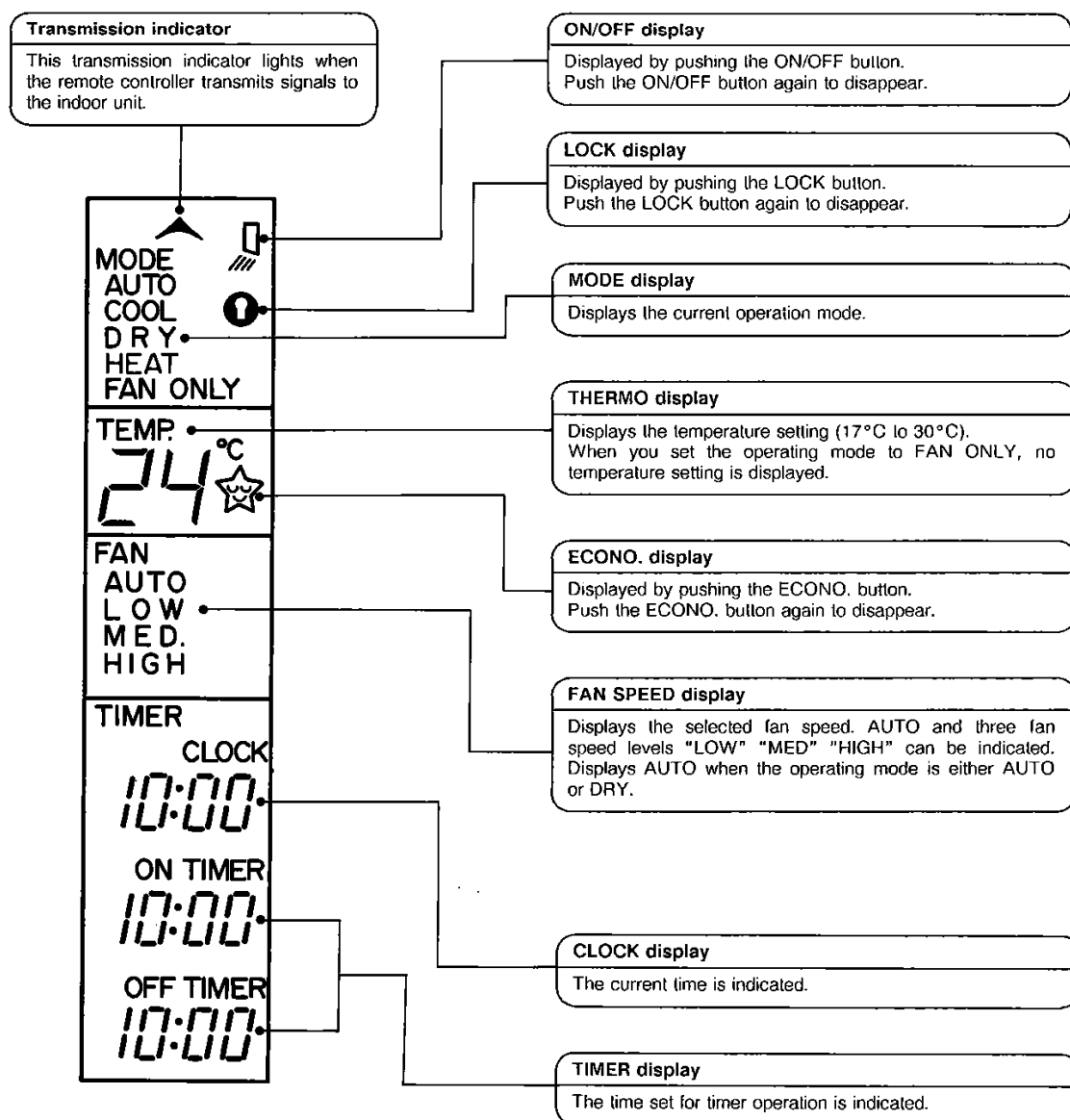
## REMOTE CONTROLS AND THEIR FUNCTIONS



## REMOTE CONTROLS AND THEIR FUNCTIONS



## NAMES AND FUNCTIONS OF INDICATORS ON REMOTE CONTROLLER

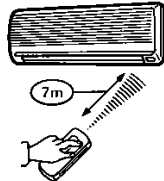


In the above illustration, all displays are indicated for the sake of clarity. During operation, only the relevant displays will be shown on the remote controller.

## HANDLING THE REMOTE CONTROLLER

### Location of the remote controller

- Keep the remote controller where its signals can reach the receiver of the air conditioner (a distance of 7m is allowed).
- When you select the timer operation, the remote controller automatically transmits a signal to the indoor unit at the specified time. If you keep the remote controller in a position that hinders proper signal transmission, a time delay of up to 15 minutes may occur.

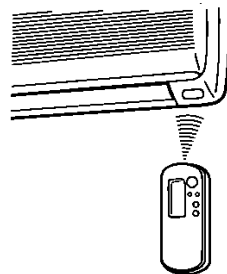


### Remote controller mounting fixture

A mounting fixture for the remote controller is supplied with it.

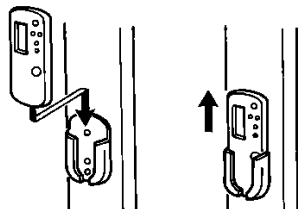
#### Installing the mounting fixture

- Before you actually install the mounting fixture on a wall or pillar, check whether the remote control unit signals from the location where you will install the fixture can be received by the indoor unit.



#### Mounting and removing the remote controller

- To mount, hold the remote controller parallel to the mounting fixture and push it in fully. To remove, slide the remote controller upwards and out from the mounting fixture.



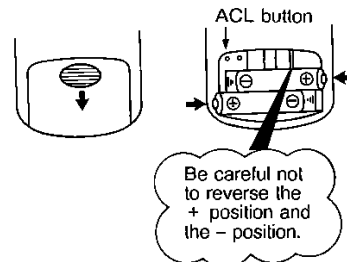
### Replacing batteries

The remote controller uses two Toshiba alkaline dry batteries (LC03).

- (1) Slide to arrow direction and remove the battery compartment cover.

Remove the used batteries and insert new batteries.

- Push the ACL button in the battery compartment.



- (2) Before replacing the battery compartment cover, confirm that the clock panel on the remote controller displays "0:00" and that the colon is flashing.

#### Notes:

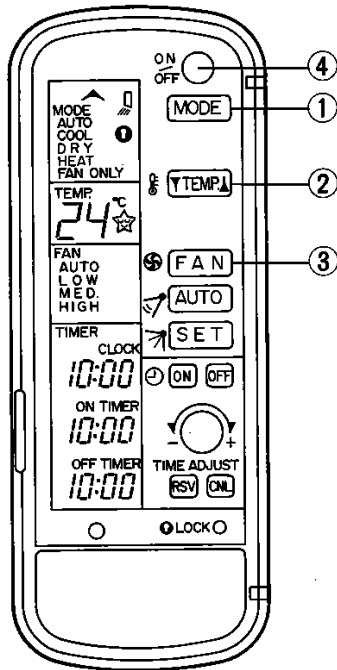
- When replacing batteries, do not use used batteries or batteries of a different type. Such use may cause the remote controller to malfunction.
- If you do not use the remote controller over a long period, remove the batteries. Otherwise, battery leakage may damage the remote controller.
- The average battery life during normal use is approximately one year.
- Replace the batteries when there is no receiving beep from the indoor unit or if the transmission indication lamp on the remote control unit fails to light.

### Cautions

- The air conditioner will not operate if curtains, doors or other materials block the signals from the remote controller to the indoor unit.
- Prevent any liquid from falling into the remote controller. Do not expose the remote controller to direct sunlight or heat.
- If the infrared signal receiver at the indoor unit is exposed to direct sunlight, the air conditioner may not function properly. Use curtains to prevent the sunlight from falling on the receiver.
- If the room using the air conditioner has fluorescent lighting with electronic starters, signals may not be properly received. If you are planning to use such fluorescent lamps, consult your local dealer.
- If other electrical appliances react to the remote control, either move these appliances or consult your local dealer.

## HEATING/COOLING/FAN ONLY OPERATION

Once you select the operating mode, the operating conditions are saved in the remote controller. Thereafter, the air conditioner will start operating under the same conditions when you simply push the ON/OFF button of the remote controller.



Start

Turn ON the POWER SUPPLY switch on the control panel of the indoor unit.  
The OPERATION lamp (green LED) on the display panel of the indoor unit starts flashing.

1

### MODE button

Select HEAT, COOL, or FAN ONLY.

2

### TEMP. button

Set the desired temperature.  
Cooling 21°C or higher.  
Heating 28°C or lower.

3

### FAN button

Select any of the "AUTO", "LOW", "MED." and "HIGH".

4

### ON/OFF button

Push this button to start the air conditioner.  
The OPERATION lamp (green LED) on the display panel of the indoor unit lights. The operating mode is selected in accordance with the room temperature and operation starts after approximately 3 minutes. (If you select FAN ONLY mode, the unit will start immediately.)

Stop

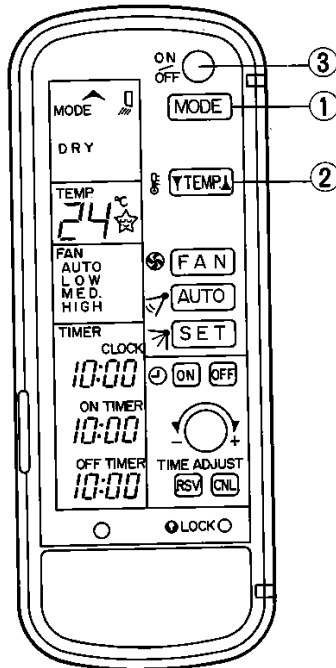
### ON/OFF button

Push this button again to stop the air conditioner.

- The FAN ONLY mode does not control temperature. Therefore, perform only steps ①, ③ and ④ to select this mode.

## DRY OPERATION

Once you select the dry operation mode, the operating conditions are saved in the unit's microcomputer memory. Thereafter, the air conditioner will start operating under the same conditions when you simply push the ON/OFF button of the remote controller.



Start

Turn ON the POWER SUPPLY switch on the control panel of the indoor unit.  
The OPERATION lamp (green LED) on the display panel of the indoor unit starts flashing.

1

**MODE button**  
Select DRY.

2

**TEMP. button**

Push the "TEMP" button.

- The fan speed indicator displays AUTO.  
The indoor fan speed will be automatically selected low.

3

**ON/OFF button**

Push this button to start the air conditioner.

- The OPERATION lamp (green LED) on the display panel of the indoor unit lights, and operation starts after approximately 3 minutes.

Stop

**ON/OFF button**

Push this button again to stop the air conditioner.

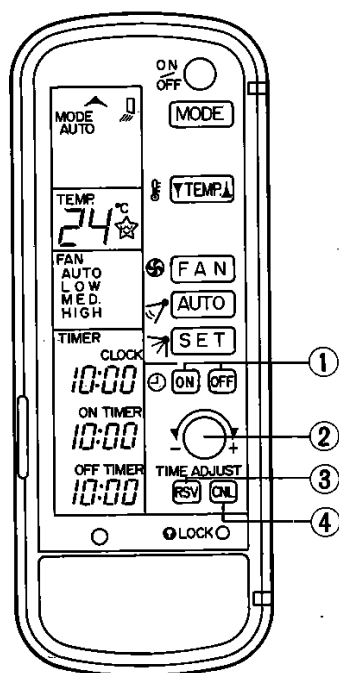
## TIMER OPERATION

Once you select the timer operation, the settings are saved in the remote controller. Thereafter, the air conditioner will start operating under the same conditions when you simply push the ON/OFF button of the remote controller.

If the clock panel displays flashing colon, you cannot set the timer.

Follow the instructions in the section

"SETTING THE CLOCK" on page 20 to set the clock, and then set the timer.



### RESET/INDICATE button

Start

Push this button. The indicators of the remote controller will display the current settings (for approximately 30 seconds).

### ON/OFF TIMER button

Push the ON or OFF TIMER button as required.

- The current timer setting is displayed along with ON/OFF TIMER indicator, and the colon in the timer setting will flash.

### Time adjust dial (TIME ADJUST)

- Turn to set the desired time.
  - Forward
  - Backward
- Each time you turn the dial, the time moves forward or backward by 10-minute depending on which side you turn. If you turn the dial quickly, the time will alter much quicker.

### Reserve button (RSV)

Push this button. Check if the TIMER lamp (yellow LED) on the display panel of the indoor unit is lit.

### Canceling

#### Cancel button (CNL)

Push the CNL button to cancel the timer setting.

### CAUTION

When you select the timer operation, the remote controller automatically transmits the timer signal to the indoor unit at the specified time. Therefore, keep the remote controller in a location from which it can transmit the signal to the indoor unit properly.

### Changing

Push the ON or OFF TIMER button once.  
The colon of the current timer setting will start flashing.  
Perform steps ② and ③ to change the setting.

## Example of Timer Setting

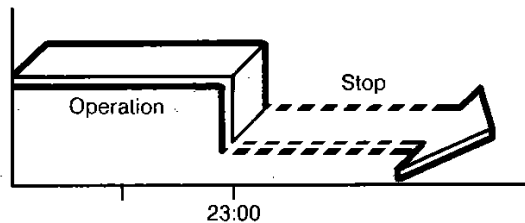
### OFF timer (Operation → Stop)

The OFF timer feature is useful when you go to sleep. The air conditioner will stop automatically at the time set.

Example:

To stop the air conditioner at 23:00 p.m.

- (1) Push the OFF TIMER button.
- (2) Use the TIME ADJUST dial to display "23:00" on the OFF timer indicator.
- (3) Push the RSV button.



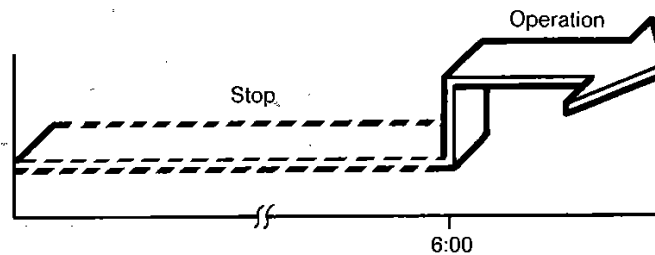
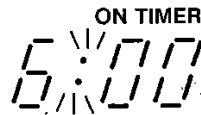
### ON timer (Stop → Operation)

The ON timer feature is useful when you wake up in the morning or when you return home. The air conditioner will automatically start operating at the time set.

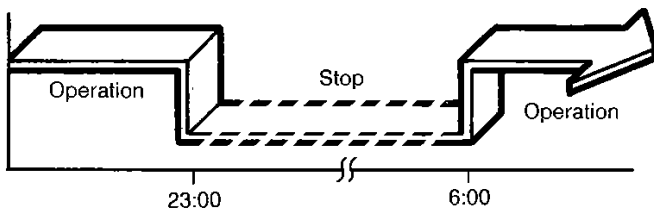
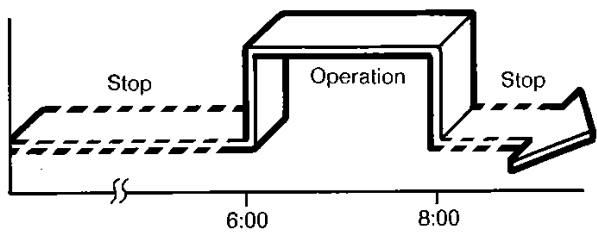
Example:

To start the air conditioner at 6:00.

- (1) Push the ON TIMER button.
- (2) Use the TIME ADJUST dial to display "6:00" on the ON timer indicator of the remote controller.
- (3) Push the RSV button.



# **Combined Timer** (Setting both ON and OFF timers simultaneously)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>OFF timer → ON timer</b><br/>(Operation → Stop → Operation)</p>                                                                                                                                                                                                                                                                                                                                                                                 | <p>This feature is useful when you want to stop the air conditioner after you go to sleep, and start it again in the morning when you wake up or you return home.</p>                     |
| <p>Example:<br/>To stop the air conditioner at 23:00 and start it again next morning at 6:00.</p> <ol style="list-style-type: none"> <li>(1) Push the OFF TIMER button.</li> <li>(2) Use the TIME ADJUST dial to display "23:00" on the OFF timer indicator.</li> <li>(3) Push the ON TIMER button.</li> <li>(4) Use the TIME ADJUST dial to display "6:00" on the ON timer indicator.</li> <li>(5) Push the RSV button.</li> </ol>                   | <div data-bbox="917 357 1177 598"> <p>OFF TIMER</p> <p>23:00</p> <p>ON TIMER</p> <p>6:00</p> </div>     |
| <p><b>ON timer → OFF timer</b><br/>(Stop → Operation → Stop)</p>                                                                                                                                                                                                                                                                                                                                                                                      | <p>You can use this setting to start the air conditioner when you wake up and stop it when you leave the house.</p>                                                                       |
| <p>Example:<br/>To start the air conditioner at 6:00 next morning, and stop it at 8:00.</p> <ol style="list-style-type: none"> <li>(1) Push the ON TIMER button.</li> <li>(2) Use the TIME ADJUST dial to display "6:00" on the ON timer indicator of the remote controller.</li> <li>(3) Push the OFF TIMER button.</li> <li>(4) Use the TIME ADJUST dial to display "8:00" on the OFF timer indicator.</li> <li>(5) Push the RSV button.</li> </ol> | <div data-bbox="803 1092 1258 1207"> <p>ON TIMER</p> <p>6:00</p> <p>OFF TIMER</p> <p>8:00</p> </div>  |

- The timer function (ON or OFF) which in time is closest to the actual time will be activated first.
- If the same time is set for both ON and OFF timers, no timer operation is performed. Also, the air conditioner may stop operating.

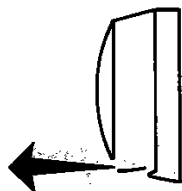
## ADJUSTING AIR FLOW DIRECTION

### Adjustment of Vertical Air Flow Direction

#### When Using the Wireless Auto Louver

##### During Operation

The air flow direction is automatically set to the desired position of the last time adjusted.



##### When You Stop Operation Using the Remote Controller

The louver is closed.



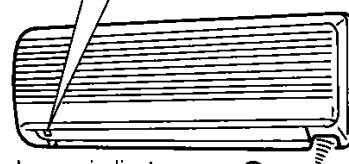
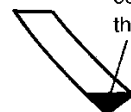
#### Setting the Air Flow Direction to Your Preference

Hold down the SET LOUVER button of the remote controller and set the air flow in the direction you desire.

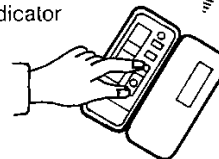
##### During Cool and Dry Operation

Change the air flow direction within the range indicated.

When the cooling or dry operation, set the louver in the blue zone.



Louver indicator



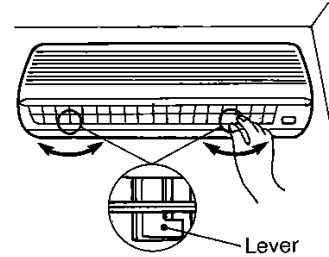
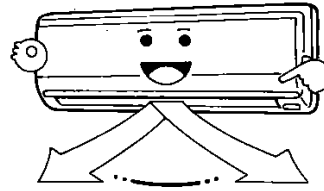
- In the Cool or Dry operation, remember not to continue the downward air flowing for a long time to avoid conceivable dewing on the louver surface and resultant dripping trouble.
- It is also possible to adjust the air flow direction by the remote controller even when the wireless auto louver is in use, but once the air conditioner operation was stopped and restarted thereafter, the air flowing direction automatically becomes horizontal.
- The air flow cannot be adjusted with the remote controller when the air conditioner is stopped (also when the ON TIMER is indicated on the display.)

### Adjusting the Horizontal Air Flow Direction

#### Preparation:

For the cooling or dry operation, move the vertical air flow louver downward using the SET button on the remote controller.

- Take hold of the lever on the horizontal air flow grilles and move them to adjust the air flow direction as required.
- You can adjust the air flow at the left, center, and right locations of the grilles.
- For the cooling or dry operation, return the vertical air flow louver to the original position using the SET button on the remote controller.

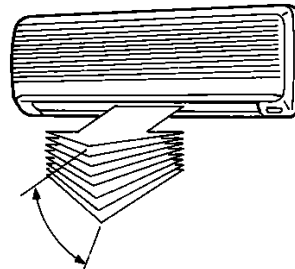


### To Automatically Swing the Air Flow Direction

Perform this function while the air conditioner is in operation.

Push the AUTO button on the remote controller.

- To stop the function, push the AUTO button again.
- To change the swing direction, push the SET button.



#### CAUTIONS

- The SET and AUTO buttons will be disabled when the air conditioner is not in operation (including when the ON TIMER is set).
- Do not operate the air conditioner for long hours with the air flow direction set downward during the cooling or dry operation. Otherwise, condensation may occur on the surface of the vertical louver and cause dew dripping.
- Do not move the vertical louver manually. Always use the SET button. If you move the louver manually, it may malfunction during operation. If the louver malfunctions, stop the air conditioner once, and restart.
- When the air conditioner is started immediately after it was stopped, the vertical louver might not move for 10 seconds or so.
- If you change the temperature setting during the heating operation (especially when the room temperature is high), air conditioner may stop, and the vertical louver may close temporarily and move to the position stored in memory.

## Operation Circuit Description

1. When the power plug (or cord) is connected to AC220/240 V (50 Hz) outlet, the operation (ON) lamp starts flashing, indicating that power has been turned on (or that power failure occurred).
2. Confirm that the TEMPORARY switch on the main unit is set to the REMOTE CONTROL position.
3. When the remote control ON/OFF button is pushed, a receive tone is heard from the main unit and the following operations are carried out.

**Fan operation** (The remote control MODE button is set to the FAN ONLY position)

Once the setting is made, the operation mode is memorized in the microcomputer so that the same operation can be effected thereafter simply by pushing the ON/OFF button.

When the FAN SPEED button is set to the AUTO position, the indoor fan motor operates as shown in Figure 1. When the FAN SPEED button is set to LOW, MED, or HIGH, the motor operates with a constant air volume as listed in Table 1. Power relay RY07 is turned off.

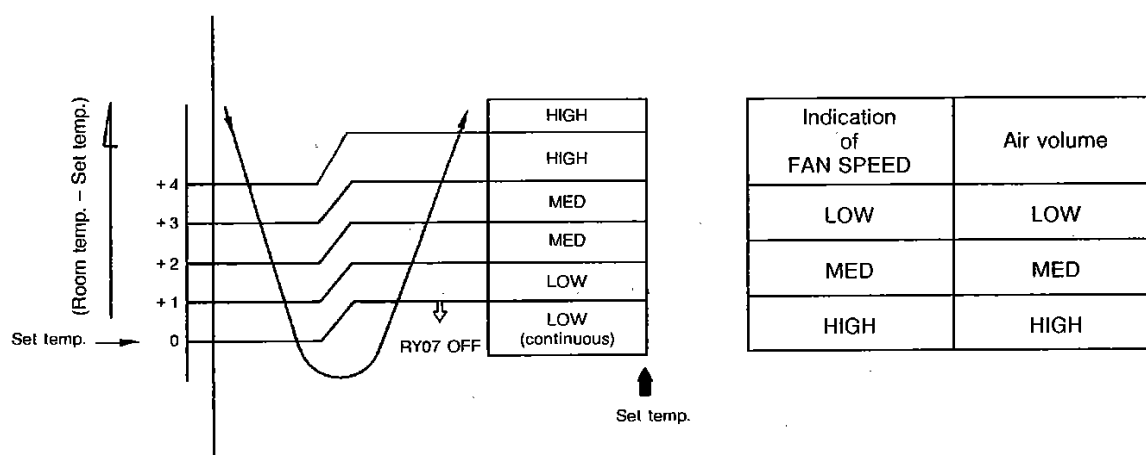


Figure 1 Auto setting of air volume

(Table 1) Manual setting of FAN SPEED

**Cool operation** (The remote control MODE button is set to the COOL position)

Once the setting is made, the operation mode is memorized in the microcomputer so that the same operation can be effected thereafter simply by pushing the ON/OFF button.

Power relay RY07 is turned on to energize the outdoor unit, and a cool operation is carried out. The indoor fan motor operates as shown in Figure 2 when FAN SPEED button is set to AUTO. The motor operates with a constant air volume as listed in Table 1, when the FAN SPEED button is set to LOW, MED, or HIGH.

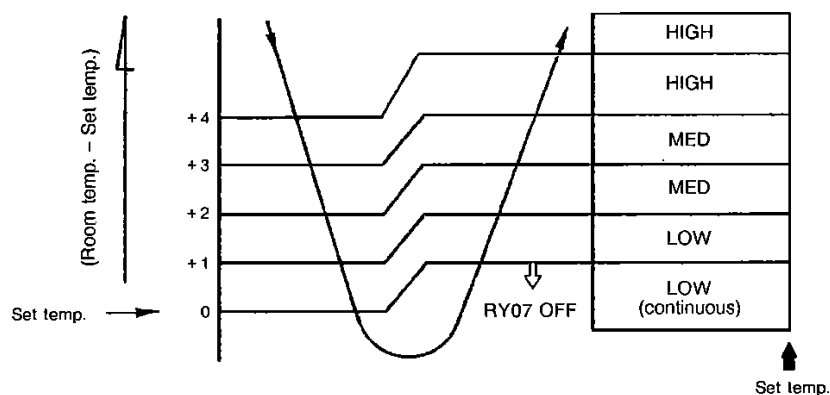


Figure 2 AUTO

a. Louver control

When the SET LOUVER button of the remote controller is pushed during the operation, the louver is automatically set to the desired position.

When the SET LOUVER button is pushed, the louver vertically swings within range of 25°.

**HEAT operation**

(The remote control MODE button is set to the Heat position)

Once the setting is made, the operation mode is memorized in the microcomputer so that the same operation can be effected thereafter simply by pushing the ON/OFF button.

Power relay RY07 is turned on to energize the outdoor unit, and a heat operation is carried out. The indoor fan motor operates as shown in Figure 3 when FAN SPEED button is set to AUTO. The motor operates with a constant air volume as listed in Table 1, when the FAN SPEED button is set to LOW, MED, or HIGH.

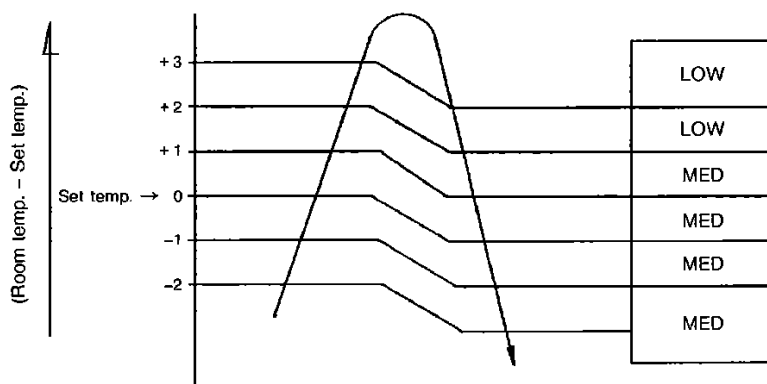
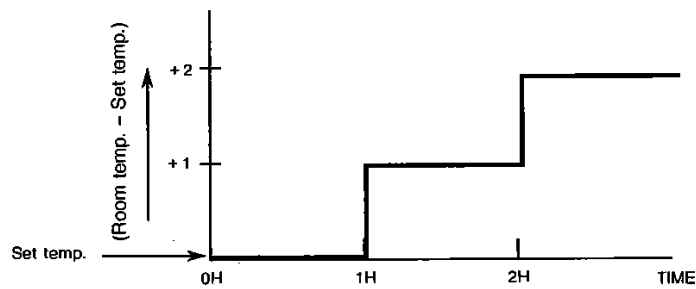


Figure 3 AUTO

**ECONO. mode**

(The remote control ECONO. button is pushed and ECONO. display is lit.)

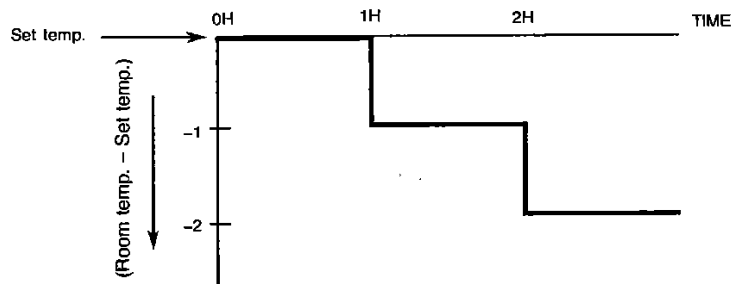
When cool operate.



When the ECONO. button is pushed, the indoor unit operates quietly and mildly with controlling airflow.

- ① In the ECONO. mode, the indoor unit will operate at set temperature for one hour from starting time, and next 1 hour, operate at temperature adding 1°C to set temperature, and then at temperature adding 2°C to set temperature since 2 hours ago.
- ② The airflow fan will operate in the Low position.

When heat operate.



When the ECONO. button is pushed, the indoor unit operates quietly and mildly with controlling airflow.

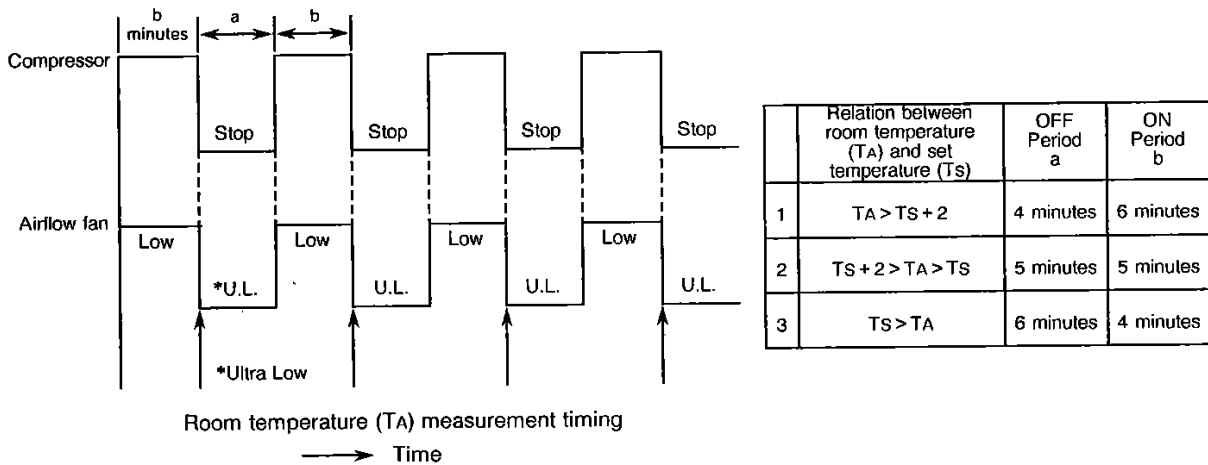
- ① In the ECONO. mode, the indoor unit will operate at set temperature for one hour from starting time, and next 1 hour, operate at temperature lowering 1°C to set temperature, and then at temperature lowering 2°C to set temperature since 2 hours ago.
- ② The airflow fan will operate in the Low position.

**Dry mode** (The remote control MODE button is set to the DRY position.)

In this mode, a microprocessor control lowers humidity while maintaining a temperature.

The microprocessor turns the compressor on and off at regular intervals (4 to 6 minutes on and/or off). During the compressor off, the airflow fan will operate in the ultra low position.

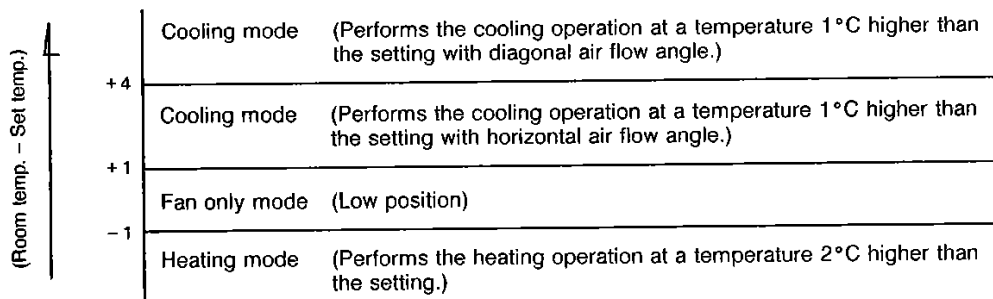
The compressor and airflow fan will operate independent of the temperature control, though the pattern of operation depends on the relation between room temperature ( $T_A$ ) and set temperature ( $T_S$ ), as shown below:



The airflow fan will operate in the Low position, independent of the position of the airflow setting control.

**Automatic operation** (The remote control MODE button is set to the AUTO position.)

One of the 3 modes, Cooling, Fan only and Heating is selected according to room temperature at which operation is to start, as shown below. The Fan mode will continue until room temperature reaches a level at which another mode is selected.



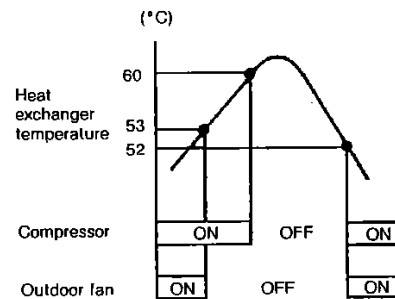
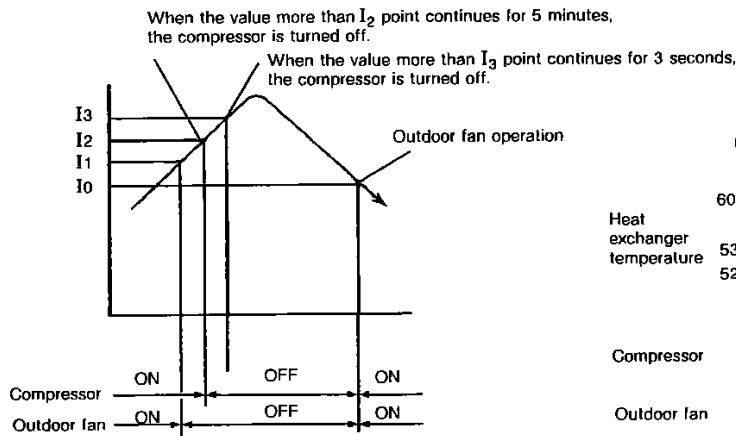
- Operation is controlled at temperatures above, independent of the position of the room temperature control.

## Current limit control and high-temperature limit control ..... (during heating)

When the load increases (overload) or the line voltage drops during operation of the air conditioner, the operating current increases, and when the power capacity is exceeded, the circuit breaker is activated. This control is performed automatically when the current (heat exchanger temperature) exceeds a certain value so that it is decreased automatically to prevent overload to the supply. Control is performed as described in ① and ②.

- ① Current limit control ..... Control is performed as shown below by detecting the compressor operation current with a current sensor.

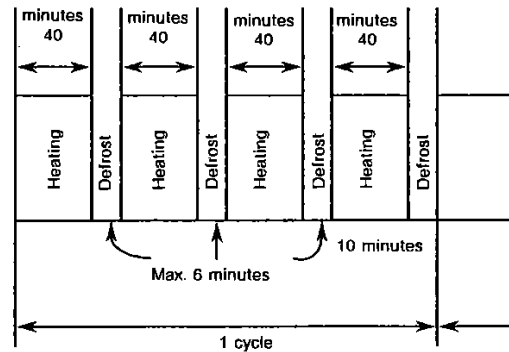
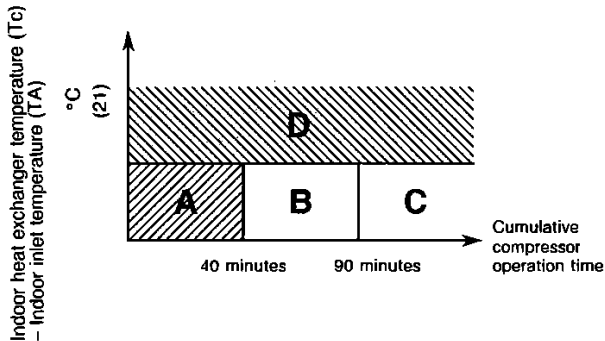
- ② High-temperature limit control ..... Control is performed as shown below by detecting the heat exchanger temperature with a heat exchanger sensor ( $T_c$ ).



## Defrost operation

- Condition to start the defrost operation

- Defrost operation time control



Defrost operation starts when the elapsed time is longer than 40 or 90 minutes, whichever specified.

**Note:** When the high-temperature limit control operation (with the outdoor fan OFF) is performed for a total of 90 minutes, defrost operation starts and continues for 10 minutes. The cumulative operation time is cleared only when the defrost operation is completed.

- The heating operation is performed for at least 40 minutes.
- The maximum defrost operation time is 6 minutes. The defrost operation time for the 4th cycle is 10 minutes. (When the outdoor temperature is very lower, however, the defrost operation time is 10 minutes.)

- Ending condition at defrost operation

When the compressor current becomes 5.5 A or more during defrost operation, the defrost operation stops and the heat operation restarts. (The current sensor detects the compressor current.)  
The defrost operation continues for at most 6 minutes.

## Defrost lamp

- During defrost operation, the PREHEAT-DEFROST lamp is on and the indoor and outdoor fans are off.
- The compressor start protection timer is interlocked with the PREHEAT-DEFROST lamp. So the PREHEAT-DEFROST lamp is off (the fans stop) for about 3 minutes after the ON/OFF button is turned on. When the compressor is turned on, the PREHEAT-DEFROST lamp comes on. After the heat exchanger is preheated to about  $30^{\circ}\text{C}$  or higher, the PREHEAT-DEFROST lamp goes off, and the indoor fan starts.

## 10. TROUBLESHOOTING CHART

| Troubleshooting procedures |                                                                                                                           |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 1                          | Check the basic items with "What to be prechecked first."                                                                 |
| 2                          | If the basic items turn out to be all right, continue troubleshooting according to "How to find faulty parts by symptom." |

### 1. WHAT TO BE PRECHECKED FIRST:

#### 1.1 Line voltage

The line voltage must be 220/240V. If the line voltage is not within this range, this air conditioner may not work normally.

#### 1.2 Incorrect cable connections between indoor and outdoor units.

The indoor unit is connected to the outdoor unit with 5 cables. Make certain that the indoor and outdoor units have been connected properly, with terminals assigned the same numbers wired to each other. If the connectors are not connected as specified, the outdoor unit will not operate normally. It should be noted that if terminals 1 and 3 or terminals 1 and 4 are connected together by mistake, some copper foil patterns on the control section PC board may be broken.

#### 1.3 Misleading but good operations (program-controlled operation)

The microcomputer performs the operations listed in the Table below to control the air conditioner. If a claim is made on the operation, check whether it corresponds to the contents in the Table. If it does, it is an indispensable operation for the control and maintenance of the air conditioner; it is not a failure of the units.

#### • Misleading but good operations

| No. | Operation of Air Conditioner                                                                                             | Description                                                                                                                                                                                                                                                                                                 |
|-----|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | When the POWER SUPPLY switch of the indoor unit is set to on, the OPERATION lamp on the setting indication part flashes. | The OPERATION lamp flashes, indicating that power is turned on. If this happens, push the ON/OFF button once to cause the lamp to stop flashing. A power outage also causes the lamp to flash.                                                                                                              |
| 2   | Fan speed remains unchanged in the dry mode.                                                                             | Fan speed is fixed at Low in the dry mode.                                                                                                                                                                                                                                                                  |
| 3   | Room temperature is in the range under which the compressor is turned off, but the compressor will not stop.             | The compressor will not stop while the compressor on-hold timer (two-minutes timer) is actuated.                                                                                                                                                                                                            |
| 4   | The compressor will not switch on or off even when the Thermo control is operated in the dry mode.                       | In the dry mode, the compressor goes on and off at regular intervals, independently of the Thermo control.                                                                                                                                                                                                  |
| 5   | The PREHEAT-DEFROST lamp comes on when the heating operation is started.                                                 | The PREHEAT-DEFROST lamp comes on during defrost operation and when the indoor heat exchanger temperature is low when the heating operation is started. At this time, the indoor fan is stopped to prevent cold air from drifting into the room.                                                            |
| 6   | The outdoor fan stops once in a while during the heating operation.                                                      | <ol style="list-style-type: none"> <li>1. When the indoor heat exchanger temperature is high, the outdoor fan may be stopped by the high-temperature limit control operation.</li> <li>2. When the compressor current is large, the outdoor fan may be stopped by the current control operation.</li> </ol> |

## 2. PRIMARY JUDGEMENT OF TROUBLE SOURCES

### 2.1 Role of indoor unit controller

The indoor unit controller receives the operation commands from the remote control and assumes the following functions.

- Measurement of the draft air temperature of the indoor heat exchanger by using the thermo sensor (TA)
- Louver motor control
- Control of the indoor fan motor operation
- Control of the LED display
- Control of the outdoor unit compressor and the outdoor fan motor

### 2.2 Display of abnormalities and judgement of the abnormal spots

The indoor unit of this machine observes the operation condition of the air conditioner and displays the contents of the self-diagnosis as block displays on the display panel of the indoor unit.

|   | BLOCK DISPLAY                               | CHECK CODE | SELF-DIAGNOSIS                                               | CHECK CODE |
|---|---------------------------------------------|------------|--------------------------------------------------------------|------------|
| A | Operation Display Flashing (1 Hz)           | —          | Power Failure (When power is on)                             | —          |
| B | Operation Display Flashing (5 Hz)           | 00         | Thermo Sensor (TA) Short/Break                               | 0C         |
| C | Operation Display Flashing (5 Hz)           | 00         | Heat Exchanger Sensor (TC) Short/Break                       | 0D         |
| D | Operation Display Flashing (5 Hz)           | 00         | Indoor Fan Lock, Abnormality of Indoor Fan, IC03 Short/Break | 11         |
| E | Operation and Timer Display Flashing (5 Hz) | 01         | Thermal Fuse is Blown (Indoor Fan Motor is overheat)         | 04         |

#### 2.2.1 Judgement from defective operation or abnormal operation

| NO. | SYSTEM                                  | CHECK                                                                                  |                                   | PRIMARY JUDGEMENT                                            |
|-----|-----------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------|--------------------------------------------------------------|
| 1.  | No reaction on remote control operation | Turn off the power once, turn it on again and try to operate the remote control again. | ① Remote control is not possible. | The Indoor part (including the remote control) is defective. |
|     |                                         |                                                                                        | ② Remote control is possible.     | O.K.                                                         |
| 2.  | The outdoor fan does not rotate         | ① The compressor operates.                                                             |                                   | The outdoor part is defective (outdoor fan motor)            |
|     |                                         | ② The compressor does not operate.                                                     |                                   | The inside part is defective.                                |

### 2.2.2 Self-diagnosis with remote controller

With the indoor unit controller, self-diagnosis of protective circuit action can be done by turning the remote controller operation into service mode, operating the remote controller, observing the remote controller indicators and checking whether ON lamp flashes (5Hz).

#### 1. How to select remote controller operation mode

##### (1) Selecting service mode

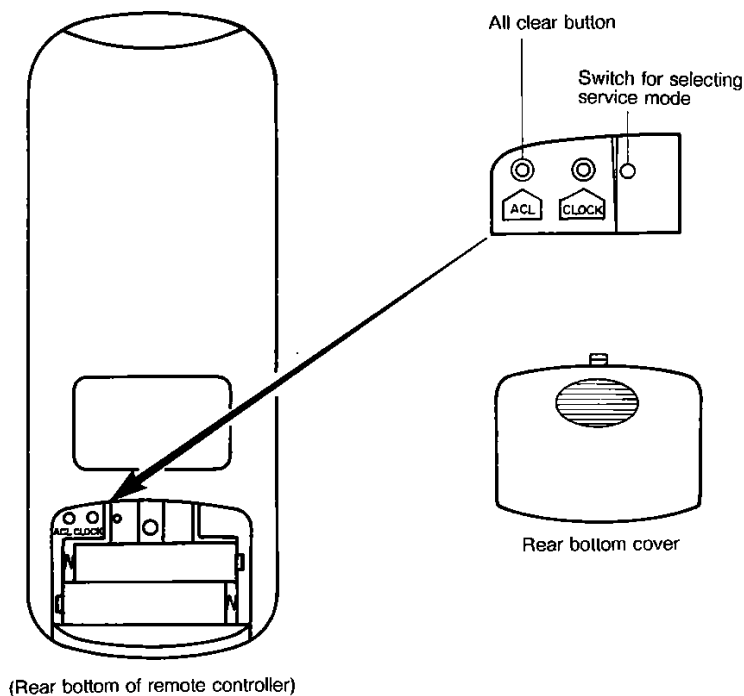
Push the switch button provided on rear bottom of the wireless remote control with a tip of pencil for more than 3 seconds. Make sure the setting temperature " " is displayed on the display and other display is turned off.

##### (2) Selecting ordinary mode

Push the all clear button (ACL) on the rear bottom of the wireless remote control with a tip of pencil for more than 3 seconds. Make sure the operation mode display, wind volume display, clock display and setting temperature display are turned on and " : " of the clock display is blinking.

#### 2. Cautions when doing service

- (1) After completion of servicing, always push the all clear (ACL) button to return the operation mode to the normal mode.
- (2) After completion of servicing by the check code, turn off the power once and then turn on the power to reset memorized contents of the microcomputer to the initial status.



### 3. Self-diagnosis by check codes

- (1) The self-diagnosis by the check codes is conducted under the block displays of item B-E.
- (2) Remote control key operation under the service mode is conducted by ON/OFF or TEMP.  
The remote control display by each key operation is varied as shown below. Two digit number is displayed in a hexadecimal number.

| Operating key                                                                                  | Indication after operation                                                  |
|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| ON/OFF                                                                                         | " 00 "                                                                      |
| TEMP.  (Up)   | 1 is added to data before operation.<br>(Example) " 02 " → " 03 "           |
| TEMP.  (Down) | 1 is subtracted from data before operation.<br>(Example) " 02 " → " 01 "    |
| "AUTO" LOUVER                                                                                  | 10 is subtracted from data before operation.<br>(Example) " 02 " → " 12 "   |
| "SET" LOUVER                                                                                   | Data before operation is directly transferred.<br>(Example) " 02 " → " 02 " |

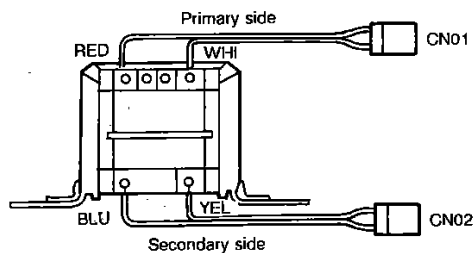
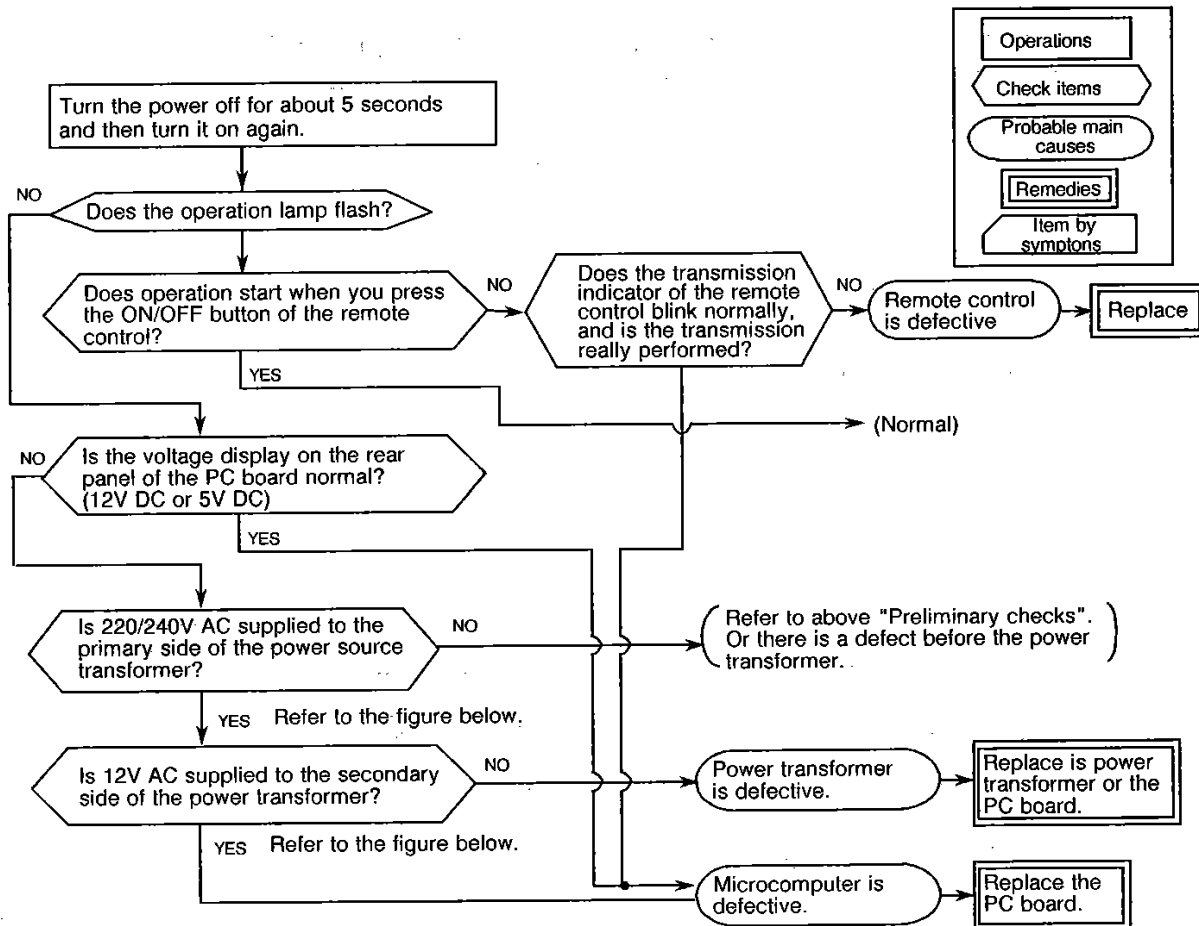
- (3) The self-diagnosis by the check codes is conducted with procedures shown below.
  - 1) Enter the service mode and make sure the off timer display of the remote control shows " 00 ".
  - 2) Operate the "ON/OFF" key and make sure the timer lamp on the display section is blinking (5Hz).
  - 3) At the same time, also make sure the operation lamp is also blinking. This shows that the protection circuit on the indoor PC board is working.
  - 4) Operate the TEMP.  key and make sure the remote control display shows " 01 " and blinking of the operation lamp. If the operation lamp is blinking, it shows the protection circuits for ????? line and serial signal system is working.
  - 5) In the same way, operate the TEMP.  key so that the display is increased one by one to continue checks by the self-diagnosis as shown in the table on P.36, 2.2. From " 00 " up to " 03 " check operations of protection circuits for each block, and " 04 " to " 1F " check operations of the typical protection circuits.

### 3. TROUBLESHOOTING FLOWCHARTS

#### 3.1 Power cannot be turned on (No operation at all)

< Preliminary checks >

- ① Is the supply voltage normal?
- ② Is the connection to the AC output O.K.?
- ③ Are the connection of the primary side and the secondary side of the power transformer inserted into the PC board?

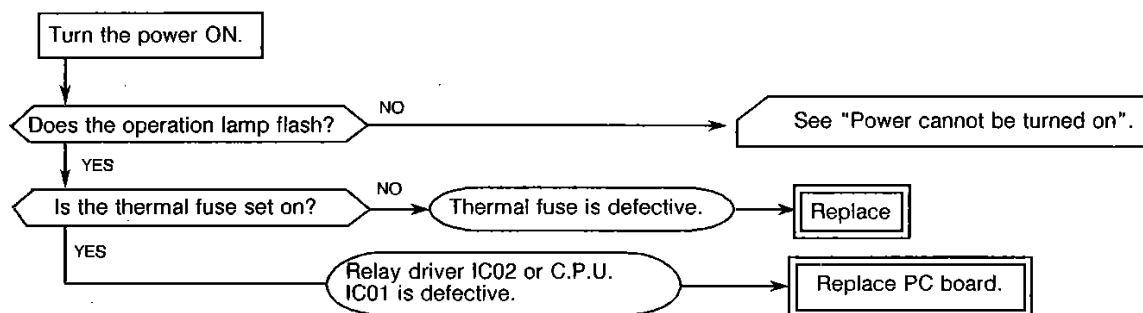


Power transformer connection diagram.

| Color Identification |        |
|----------------------|--------|
| WHI:                 | WHITE  |
| RED:                 | RED    |
| YEL:                 | YELLOW |
| BLU:                 | BLUE   |

### 3.2 Power relay RY7 does not operate.

Louver is not controlled automatically.



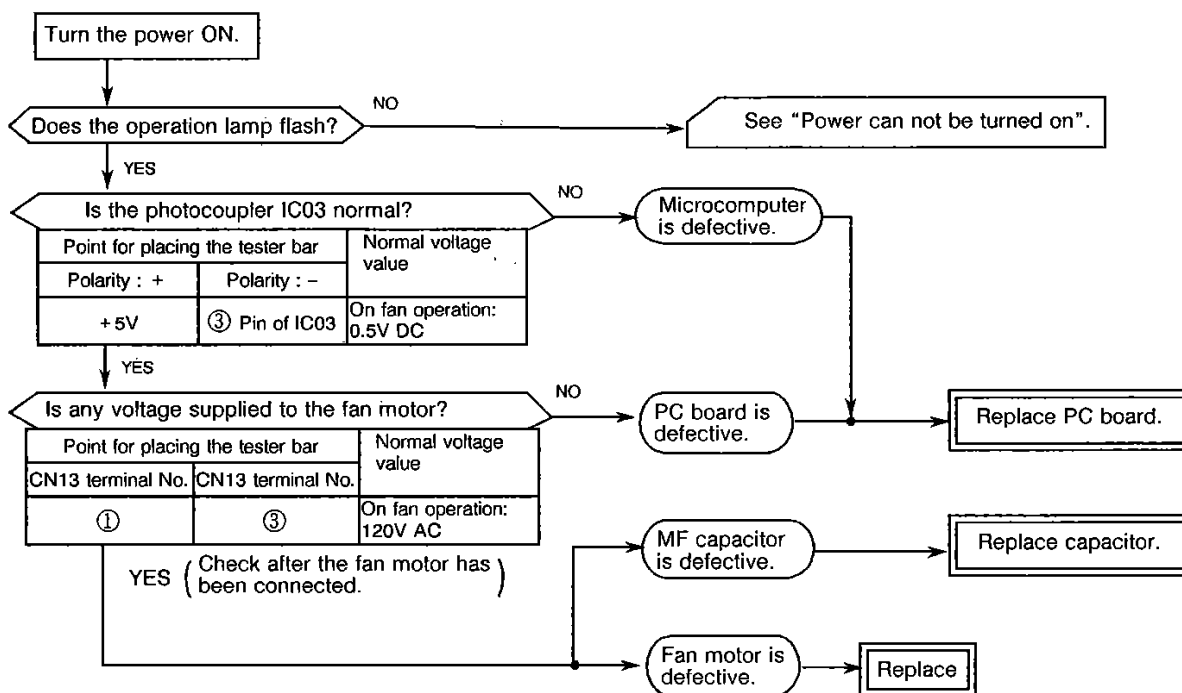
●When wiring to the thermal fuse has been broken, the timer lamp and operation lamp will flash with 5 Hz.

### 3.3 Only the indoor fan does not operate

< Preliminary checks >

Does it neither work in COOL or FAN ONLY operation?

< Checking procedure >

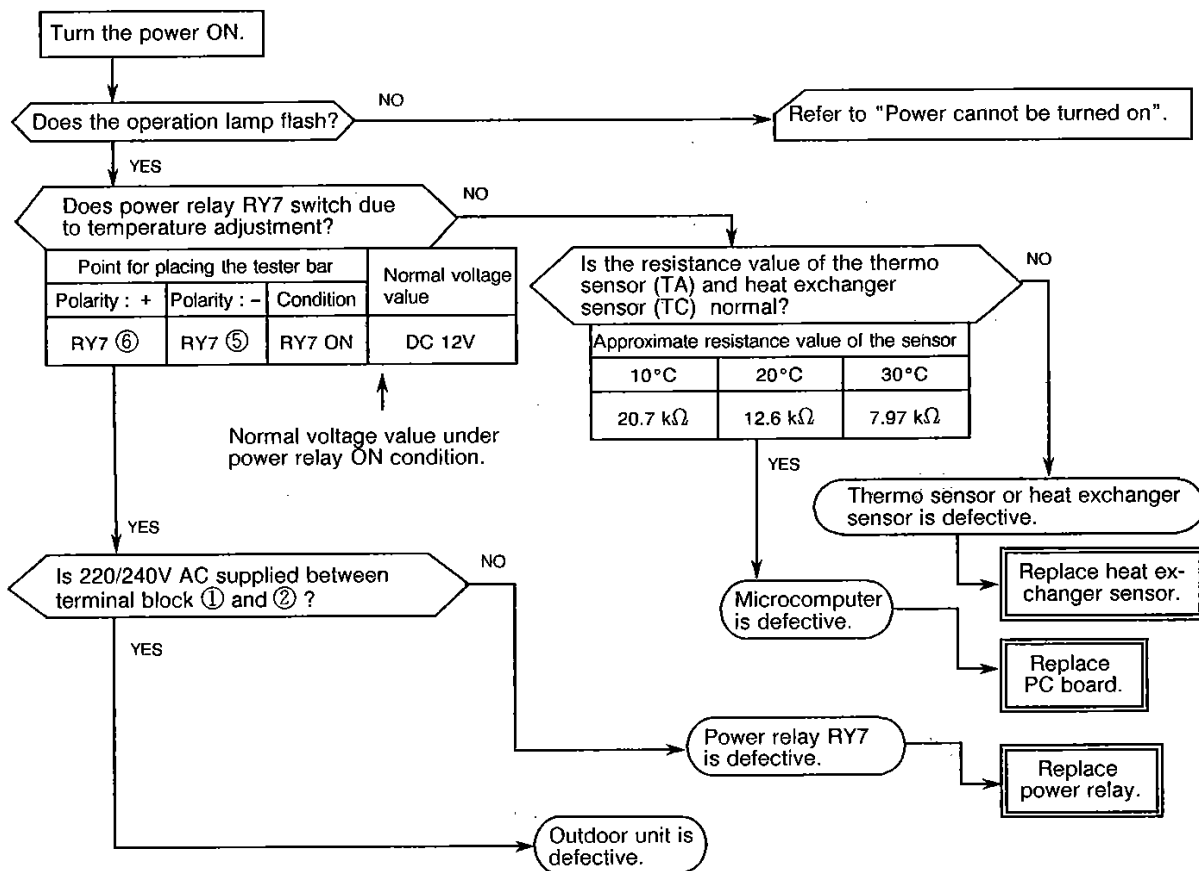


### 3.4 Compressor does not operate (Indoor fan also does not turn.)

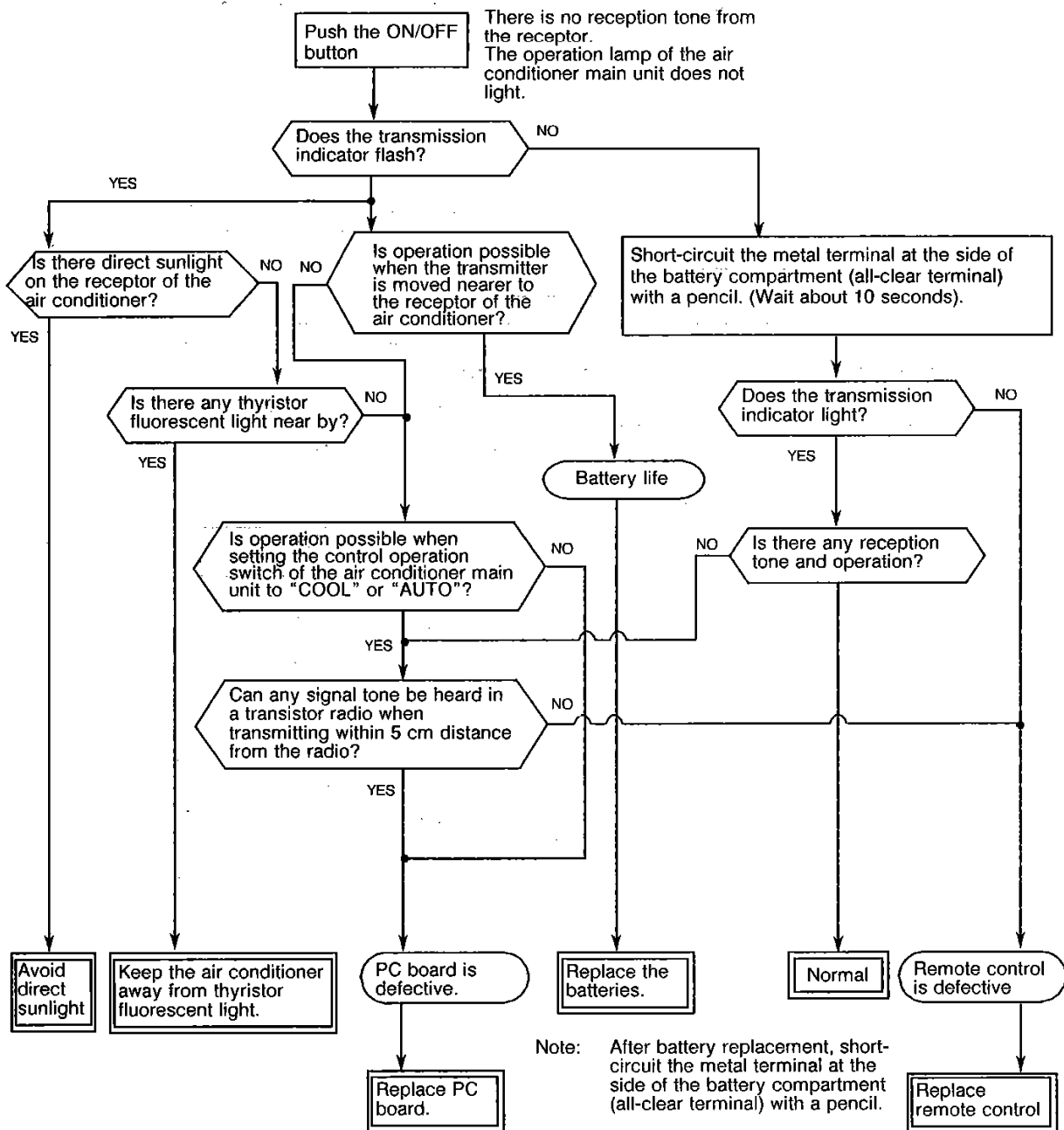
< Preliminary checks >

- ① Is the temperature set on the remote control higher than the room temperature in cool operation?
- ② Is contact of the crossing wiring O.K.?

< Checking procedure >



## 5. HOW TO CHECK THE REMOTE CONTROL (INCLUDING THE INDOOR PC BOARD)



## 5.1 HOW TO CHECK THE PC BOARD

### 5.1.1 Operating precautions

- ① When removing the front panel or the PC board, be sure to disconnect the power plug from the AC outlet.
- ② When removing the PC board, hold the edge of the PC board and do not apply force to the parts.
- ③ When connecting or disconnecting the connectors on the PC board, hold the whole housing. Do not pull at the lead wire.

### 5.1.2 Inspection procedures

- ① When a PC board is judged to be defective, check for disconnection, burning, or discoloration of the copper foil pattern or this PC board.
- ② The PC board consists of the following 4 parts:
  1. Main PC board part..... power relay, indoor fan motor drive circuit and control circuit, C.P.U. and peripheral circuits, buzzer drive circuit and buzzer.
  2. Light receiving unit..... light receiving circuit
  3. Display ..... LED
  4. Switch PC board ..... wireless-control, TEMPORARY switch
 Check the defects of the PC board following the list below.  
 (Referring to Table-4)

### 5.1.3 Checking procedure

| No. | Procedure                                                                                                                                                       | Check point (Symptom)                                                                                                                                                                                                              | Trouble cause                                                                                                                                                                                               |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ①   | Disconnect the power plug from the AC outlet and remove the PC board assembly from the electronic parts base.<br>Remove the flat cable from the terminal plate. | Is the fuse blown?                                                                                                                                                                                                                 | ① Application of shock voltage<br>② Short-circuit of the indoor fan motor                                                                                                                                   |
| ②   | Turn the power ON.<br>If the operation lamp flash (0.5 sec. ON, 0.5 sec. OFF), steps ①–③ in the right column are not necessary.                                 | Check power supply voltage.<br>① Between CN01 ① and ③ (220/240V AC)<br>② Between CN02 ① and ③ (12V AC)<br>③ Between Q01 emitter and GND (5V DC)<br>④ Between Q03 emitter and GND (12V DC)<br>⑤ Between CN11 ① pin and GND (12V DC) | ① Defective power cord, power switch, fuse or line filter, or wrong wiring<br>② Defective power transformer<br>③ Defective power circuit or short-circuited load<br>④ Same as ③<br>⑤ Thermal fuse operation |
| ③   | Push the ON/OFF button once to set in operation mode.<br>(Do not set to the fan only or on-timer mode.)                                                         | Check power supply voltage.<br>① Power relay coil voltage (12V DC) (RY7 ⑤,⑥)<br>② Between PF terminals 1 and 2                                                                                                                     | ① Relay coil cable is broken, relay driver (IC181) is defective.<br>② Relay contact is defective, PF terminal is defective.                                                                                 |
| ④   | Start operation by using the anti-restart timer.                                                                                                                | ① All LEDs of the operation lamp, the timer lamp and the FAN ONLY lamp light up.<br>② After 3 seconds, normal display does not appear.                                                                                             | } Display is defective or defect in the 4P housing assembly.                                                                                                                                                |

| No. | Procedure                                                                                                                                                                                                                                    | Check point (Symptom)                                                                                                                                                                                                  | Trouble cause                                                                                                       |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| ⑤   | Push the ON/OFF button once to set in operation mode.<br>1. Setting the anti-restart timer<br>2. Cooling operation<br>3. Fan speed : AUTO<br>4. Set the temperature sufficiently lower than the room temperature.<br>5. Continuous operation | ① The compressor does not operate.<br>② The operation lamp flashes.                                                                                                                                                    | ① The temperature of the indoor heat exchange unit is extremely low.<br>② Defective control PC board.               |
| ⑥   | Start operation as follows:<br>1. Set the operation mode to "FAN ONLY".<br>2. Set the fan speed to "HIGH".<br>3. Continuous operation                                                                                                        | ① There is a voltage of 120V or more between the red and black motor connector leads.<br>② The motor does not rotate. (But the key operation of the remote control is accepted.)<br>③ Motor rotates but vibrates hard. | ① Indoor fan motor is defective.<br>② Contact of the motor connector is defective.<br>③ Main PC board is defective. |

Approximate value of the sensor (thermistor) resistance  
(TA, TC)

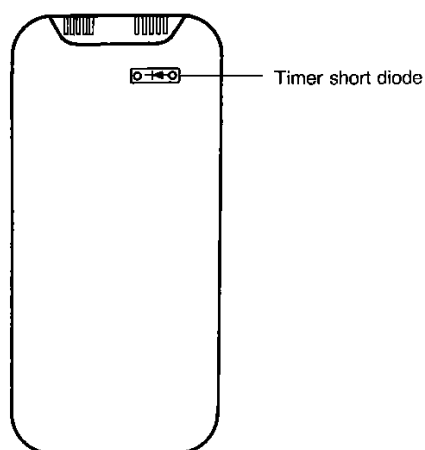
( $\approx$  k $\Omega$ )

| Sensor \ Temperature | 0°C  | 10°C | 20°C | 30°C |
|----------------------|------|------|------|------|
| Thermo Sensor        | 35.8 | 20.7 | 12.6 | 7.92 |

## 5.2 How to reduce the operation time of the anti-restart timer

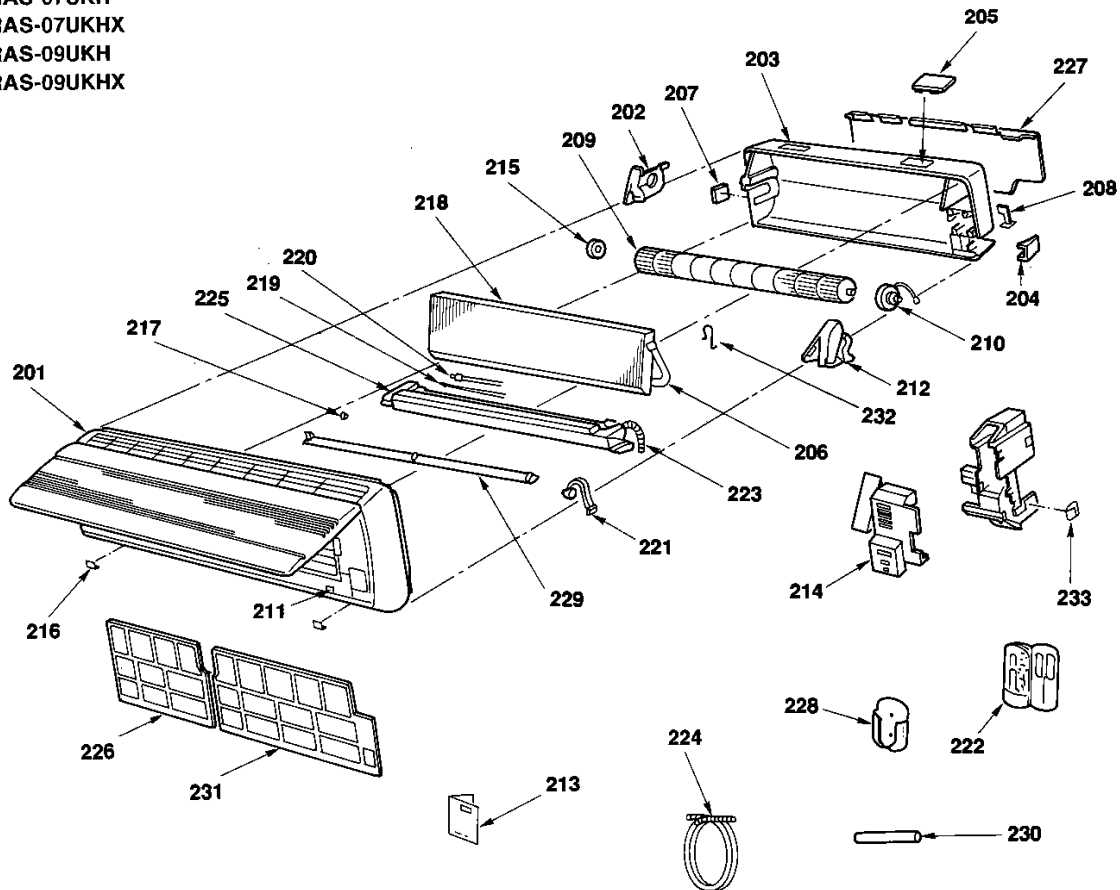
- Drill 2 holes on the rear of the wireless remote control unit.  
Attach the diode (IC1555 or equivalent) to the rivet inside the unit.
- Push the ON/OFF button to start operation with the diode attached.  
During operation, push the transmission/check button while the diode is attached.

Wireless remote controller



## 11. EXPLODED VIEWS AND PARTS LIST

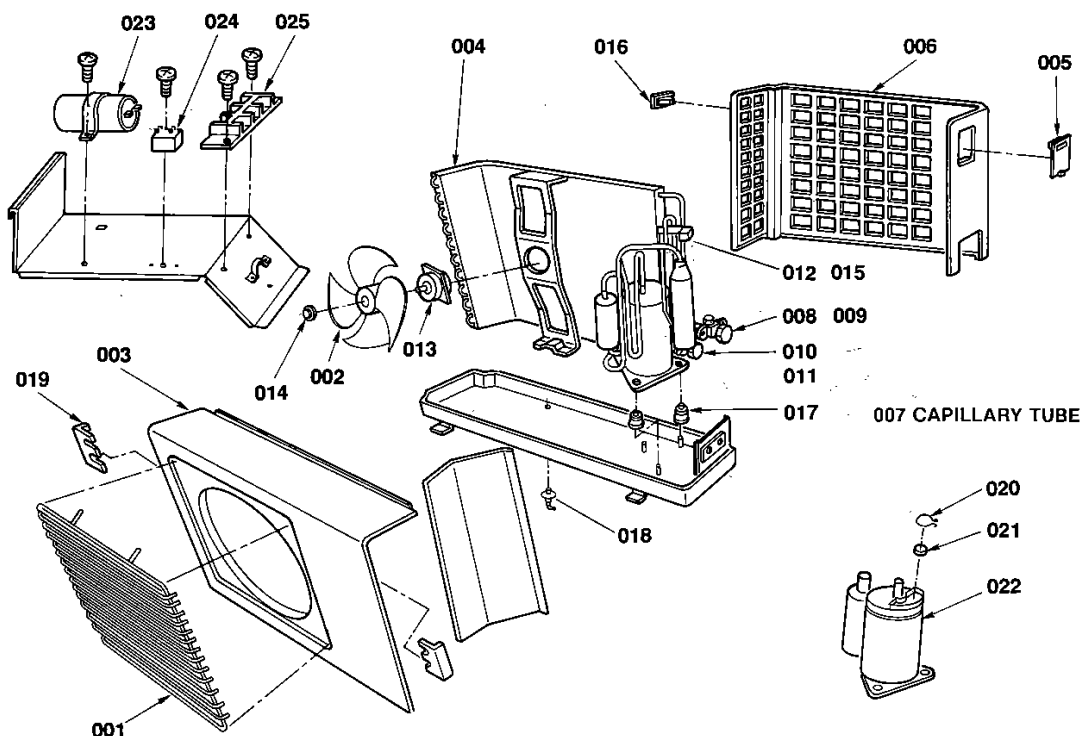
11.1 Indoor unit  
RAS-07UKH  
RAS-07UKHX  
RAS-09UKH  
RAS-09UKHX



| Location No. | Part No. | Description                                                |
|--------------|----------|------------------------------------------------------------|
| 201          | 43005115 | Front Panel (09UKH)                                        |
| 202          | 43039251 | Base Bearing                                               |
| 203          | 43003218 | Rear Plate                                                 |
| 204          | 43007963 | Bush, Rear Plate, Right                                    |
| 205          | 43007967 | Lid, Rear Plate                                            |
| 206          | 43011503 | Pipe, Sealed                                               |
| 207          | 43007964 | Bush, Rear Plate, Left                                     |
| 208          | 43019836 | Holder, Pipe                                               |
| 209          | 43020244 | Fan, Cross-Flow                                            |
| 210          | 43021907 | Motor, Fan, AC200V 50Hz                                    |
| 211          | 43007842 | Latch                                                      |
| 212          | 43039250 | Motor Band, Left                                           |
| 213          | 43088997 | Owner's Manual (English/French/<br>German/Spanish/Italian) |
| 214          | 43006012 | Cover, E-Parts                                             |
| 215          | 43022321 | Bearing                                                    |
| 216          | 43007961 | Cap, Screw                                                 |

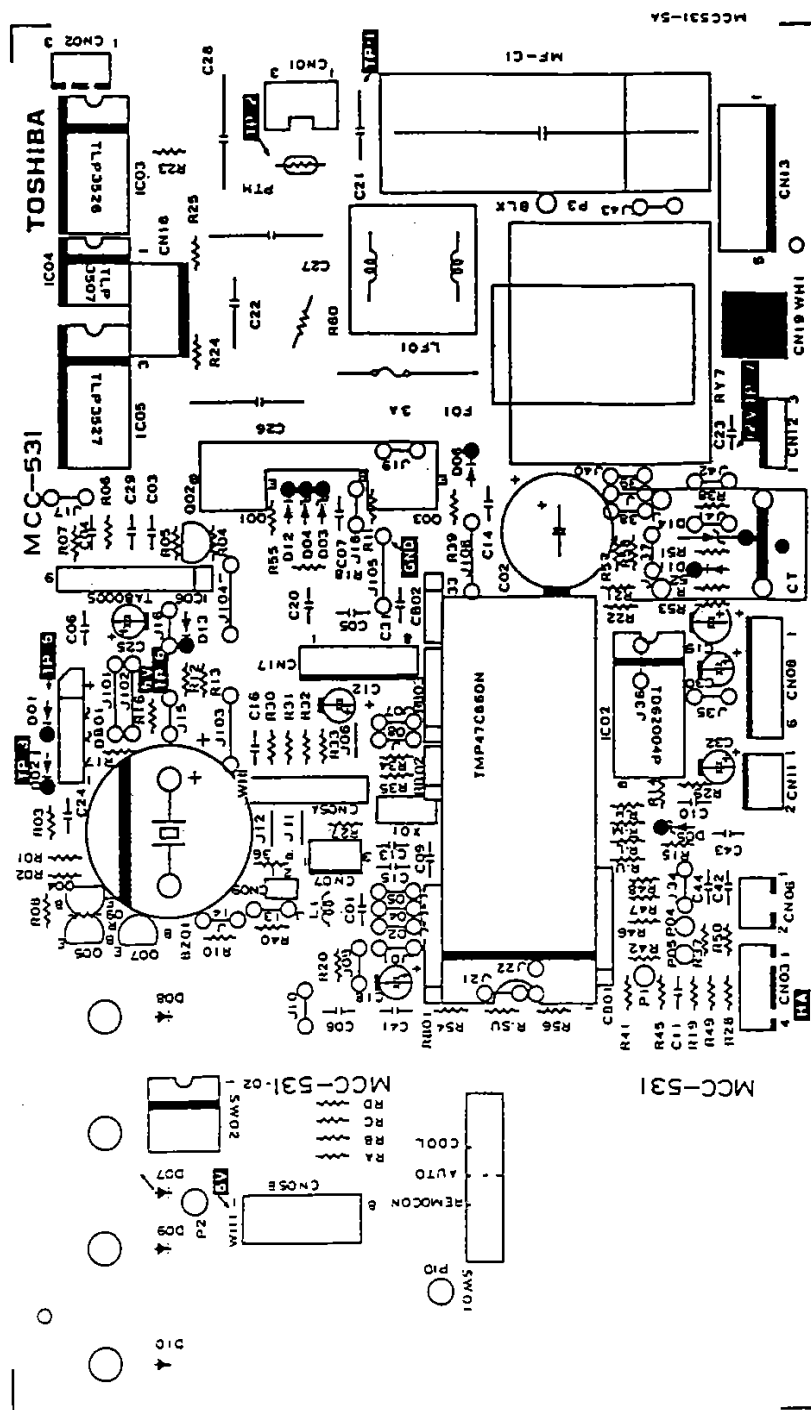
| Location No. | Part No. | Description               |
|--------------|----------|---------------------------|
| 217          | 43007843 | Bushing                   |
| 218          | 43044559 | Evaporator                |
| 219          | 43047331 | Pipe, Delivery            |
| 220          | 43047332 | Pipe, Suction             |
| 221          | 43021794 | Motor, Louver             |
| 222          | 43069666 | Remote Controller         |
| 223          | 43070140 | Hose, Drain               |
| 224          | 43070090 | Hose, 3M, Drain           |
| 225          | 43009450 | Grille A'ssy              |
| 226          | 43080359 | Air Filter (Left)         |
| 227          | 43082256 | Plate Inst                |
| 228          | 43063253 | Holder, Remote Controller |
| 229          | 43009408 | Grille                    |
| 230          | 43011499 | Pipe, Sealed              |
| 231          | 43080358 | Air Filter (Right)        |
| 232          | 43019822 | Fix-P-Sensor              |
| 233          | 43007039 | Cover, Terminal           |

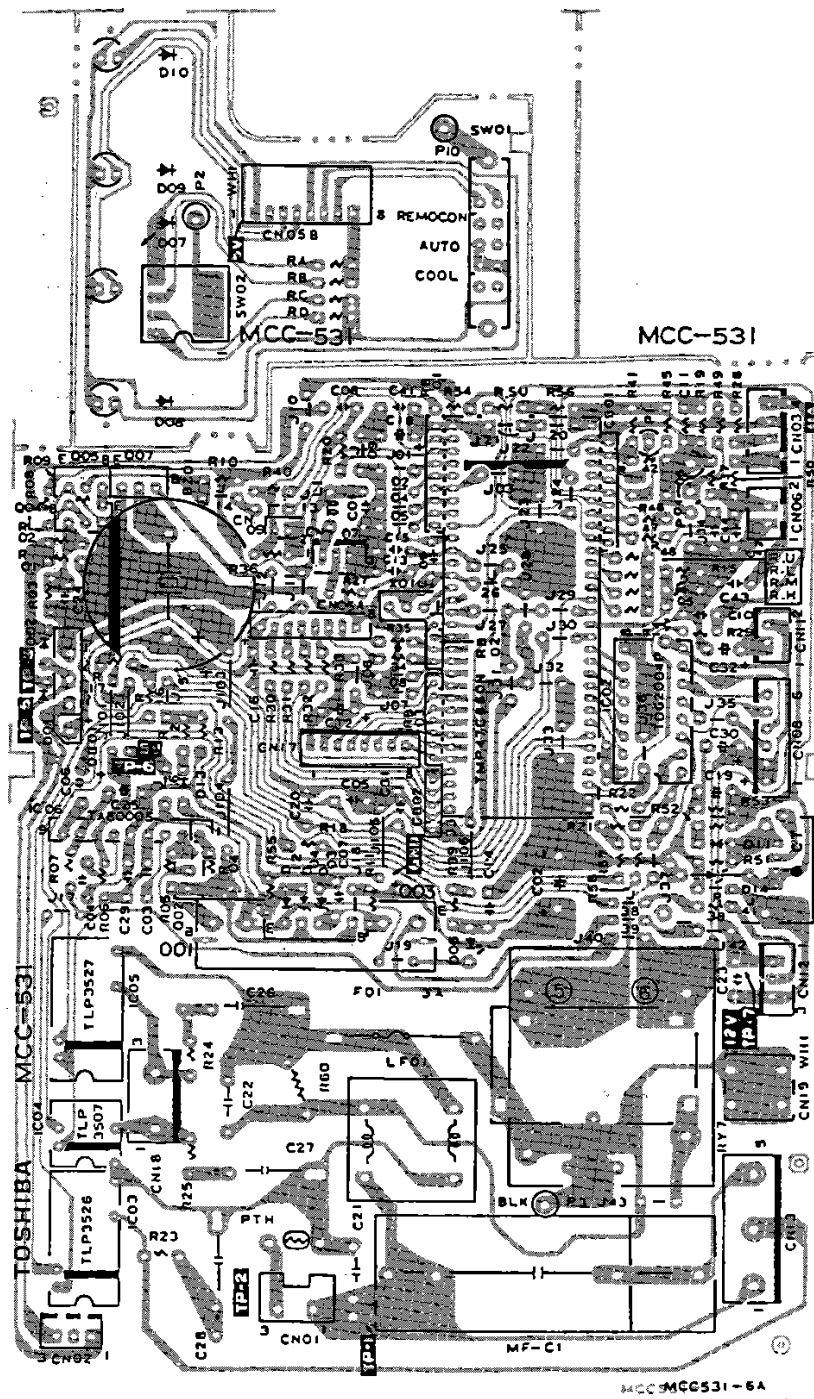
11.2 Outdoor unit  
RAS-07UAH  
RAS-07UAHX  
RAS-09UAH  
RAS-09UAHX

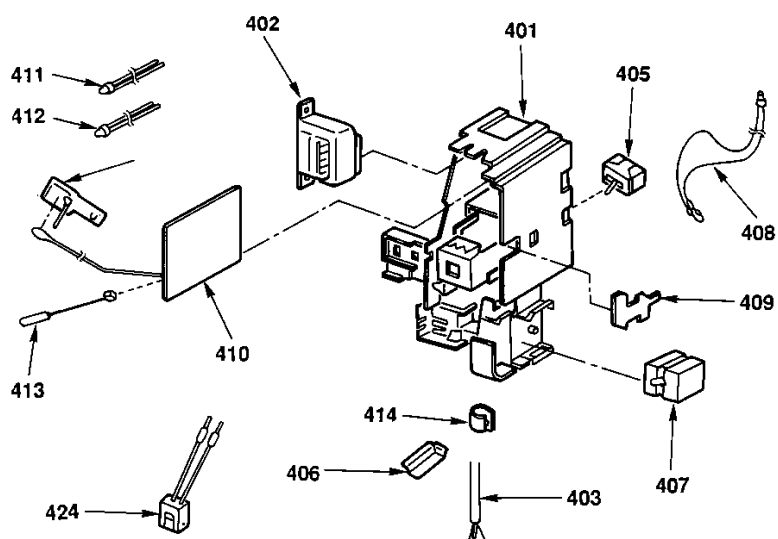


| Location No. | Part No. | Description                                                      |
|--------------|----------|------------------------------------------------------------------|
| 001          | 43019769 | Guard, Fan                                                       |
| 002          | 43020223 | Fan, Propeller                                                   |
| 003          | 43005037 | Cabinet, Front                                                   |
| 004          | 43043548 | Condenser                                                        |
| 005          | 43062124 | Cover, E-Parts                                                   |
| 006          | 43005038 | Cabinet, Back                                                    |
| 007          | 43047491 | Capillary Tube<br>(07UAH,X ø1.5x1700 ℓ)<br>(09UAH,X ø1.5x1500 ℓ) |
| 008          | 43046229 | Packed Valve, 3/8                                                |
| 009          | 43047401 | Bonnet, 3/8                                                      |
| 010          | 43046228 | Packed Valve, 1/4                                                |
| 011          | 43147196 | Bonnet, 1/4                                                      |
| 012          | 43046248 | 4-Way Valve, VH7100D                                             |
| 013          | 43021928 | Fan-Motor, UE6-21A5P                                             |
| 014          | 43047549 | Flange, Nut                                                      |
| 015          | 43046255 | Solenoid Coil                                                    |

| Location No. | Part No. | Description                                  |
|--------------|----------|----------------------------------------------|
| 016          | 43019727 | Handle                                       |
| 017          | 43049643 | Rubber-Cushion                               |
| 018          | 43032441 | Nipple, Drain                                |
| 019          | 43019770 | Stopper, Guard                               |
| 020          | 43069988 | Holder, OL-Relay                             |
| 021          | 43054313 | Relay, Overload<br>(RAS-07UAH, 07UAHX)       |
| 021          | 43054391 | Relay, Overload<br>(RAS-09UAH, 09UAHX)       |
| 022          | 43041142 | Compressor, PH70T1-4C<br>(RAS-07UAH, 07UAHX) |
| 022          | 43041564 | Compressor, PH94T1-4C<br>(RAS-09UAH, 09UAHX) |
| 023          | 43055365 | Capacitor, Plastic Film, 20MFD, 400V         |
| 024          | 43055409 | Capacitor, 1.5MFD, 400V                      |
| 025          | 43060977 | Terminal-5P                                  |

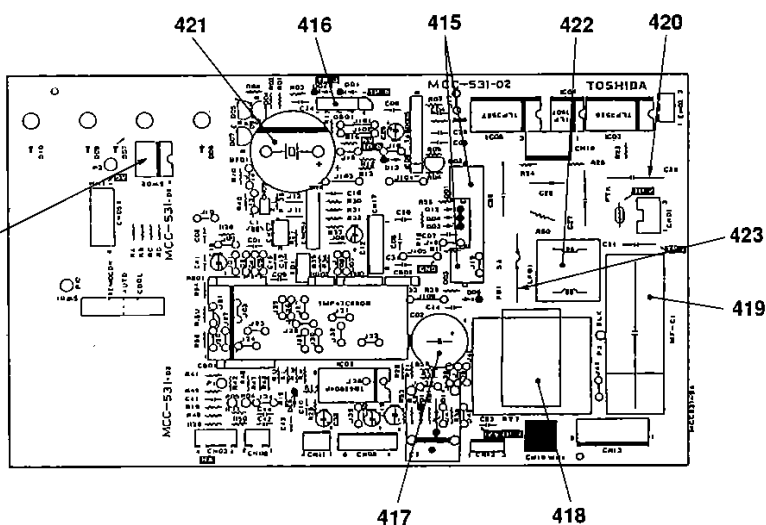






SW02 is to be set to the position below.

| Model \ SW02 bit No. | 1   | 2  | 3   | 4  |
|----------------------|-----|----|-----|----|
| 07PKH                | OFF | ON | OFF | ON |
| 09PKH                | OFF | ON | OFF | ON |



| Location No. | Part No. | Description                 |
|--------------|----------|-----------------------------|
| 401          | 43061177 | Base, E-Parts               |
| 402          | 43058241 | Transformer, Power, ST-8916 |
| 403          | 43060982 | Power Cord                  |
| 405          | 43051315 | Switch, Power               |
| 406          | 43069573 | Receiver, Infrared Rays     |
| 407          | 43060778 | Terminal Block              |
| 408          | 43060748 | Set-Micro-Temp              |
| 409          | 43063257 | Holder - P.C. Board         |
| 410          | 43068189 | P.C. Board Ass'y            |
| 411          | 43035492 | Housing-Assemble-Primary    |
| 412          | 43035466 | Housing Ass'y               |
| 413          | 43068011 | Sensor, Heat Exchanger      |

| Location No. | Part No. | Description                        |
|--------------|----------|------------------------------------|
| 414          | 43063181 | Clamp, Cord                        |
| 415          | 43031043 | TR 2SD880Y                         |
| 416          | 43031069 | Diode, Block                       |
| 417          | 43033182 | Capacitor, Electrolytic            |
| 418          | 43034017 | Relay, Power                       |
| 419          | 43055412 | Capacitor, 1.0MFD, 400V            |
| 420          | 43033181 | Capacitor, Plastic                 |
| 421          | 43035327 | Buzzer, Piezoceramic               |
| 422          | 43034046 | Filter, Line                       |
| 423          | 43060884 | Fuse, 3A, 250V                     |
| 424          | 43035527 | Housing Assembly, 4-Way, Fan, Lead |